



Vlaanderen
is supercomputing

VSC HPC introduction course

<https://hpcleuven.github.io/HPC-intro>



VLAAMS
SUPERCOMPUTER
CENTRUM

*Innovative Computing
for A Smarter Flanders*

Training material

- ❑ Everything is available on Github:
<https://hpcleuven.github.io/HPC-intro/>
- ❑ Video recordings
 - Scan the QR code
 - Recommended videos: ~2 hrs
 - Optional videos: ~1 hr

By registering and participating in VSC trainings, you accept to abide by the [VSC training Code of Conduct](#)



What is High Performance Computing?

Concept

Using many computers together to solve large-scale, complex problems faster than one computer can

Hardware Layer

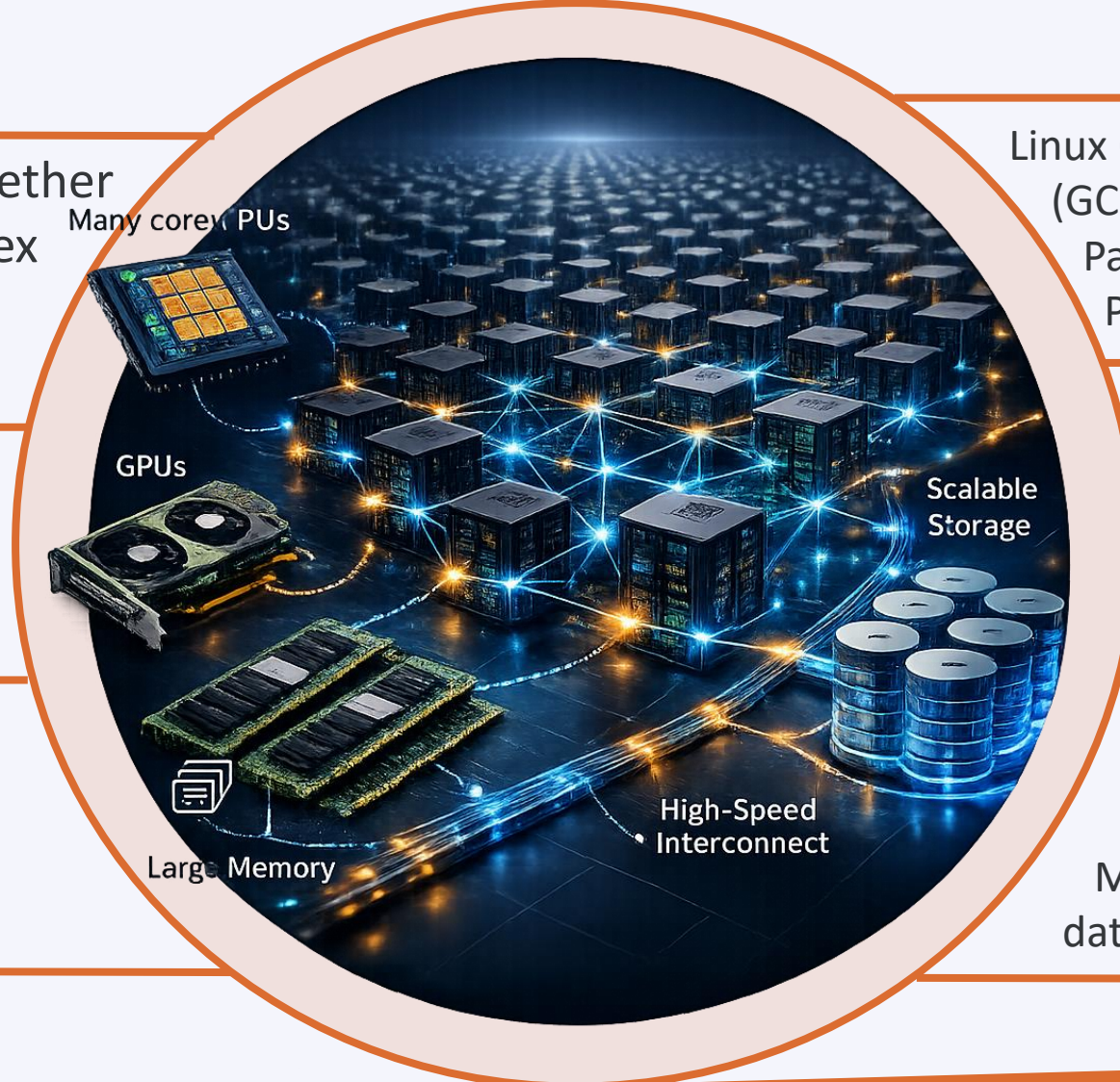
Hundreds of nodes
Tens of cores per node
Large memory
Fast interconnect
GPUs and Network filesystem

Software Layer

Linux OS + system libraries
(GCC) Compilers
Parallel libraries (OpenMP, MPI, ...)
Parallelized software

Applications

Genomics
Numerical (CFD) simulations
Molecular dynamics
Long-term weather predictions
Machine learning and AI
data pre-/post-processing



What is High Performance Computing?



The concept is simple: **parallelism** = employing multiple processors for a single problem

Training outline

Part 1: Basics

- ☐ VSC
- ☐ Tier-2 clusters
- ☐ Storage
- ☐ Accounting
- ☐ Login
 - Open OnDemand
 - SSH client & terminal
- ☐ Data transfer
 - FileZilla
 - rsync
 - Globus

Part 2: Hands-on

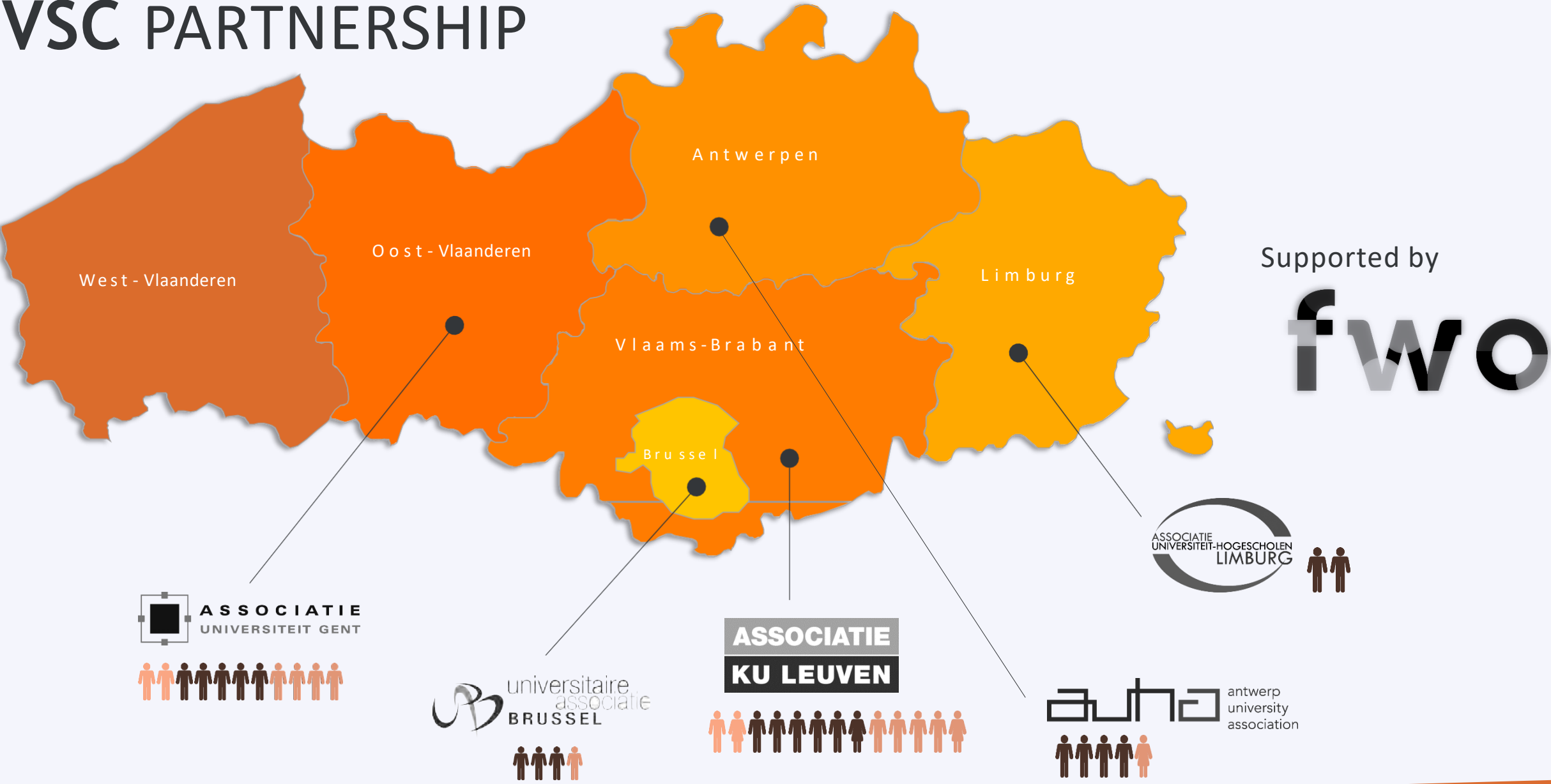
- ☐ Using Open OnDemand
- ☐ Software
- ☐ Miniconda
- ☐ Batch jobs
- ☐ Demo: try things yourself



VSC

(Vlaams Supercomputer Centrum)

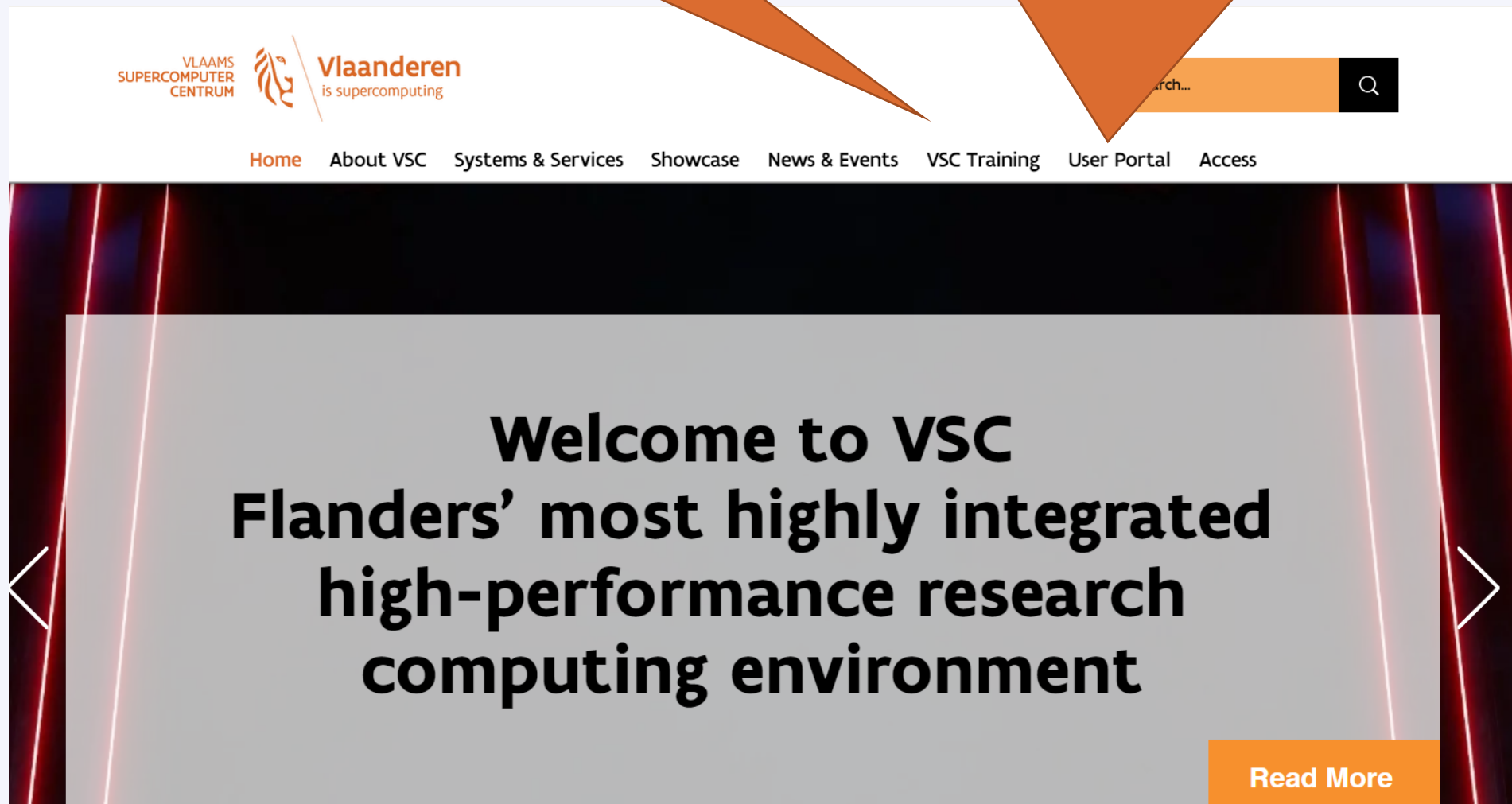
VSC PARTNERSHIP



Visit <https://vscentrum.be>

Extensive set of courses
every academic year

Links to the VSC **account page** and the VSC **documentation**
(answers >80% of your questions!)





Welcome to the User Portal

Here you can find the gateway to the User documentation of the Vlaams Supercomputer Centrum

[User documentation page](#)

Manage your VSC-account

partner institute account required

[VSC account page](#)

<https://docs.vscentrum.be>

The screenshot shows a web browser window with the URL <https://docs.vscentrum.be>. The page header includes the VLAAMS SUPERCOMPUTER CENTRUM logo and navigation links: About, Contact, Accounts, Compute, Data, Cloud, and FAQs. A search icon is located in the top right corner of the header. An orange callout box with an arrow points to this search icon, containing the text: "Enter your keywords here; e.g. 'OnDemand'". The main content area has a dark background and features a large heading "Welcome to the VSC documentation". Below this, a paragraph states: "The VSC documentation offers extensive *how-to* guides and technical information about the services provided by the Vlaams Supercomputer Centrum." The page is organized into three main sections: "VSC Accounts", "Compute", "Data", and "Cloud". The "VSC Accounts" section includes a sub-header "How to get your VSC account to use the different VSC services and platforms." The "Compute" section describes the high-performance computing (HPC) platform and lists "Tier-1 HPC", "Tier-2 HPC", and "Terminal Interface". The "Data" section describes the VSC Data component and lists "Tier-1 Data". The "Cloud" section describes the VSC Cloud component and lists "Virtual Machines".

Welcome to the VSC documentation

The VSC documentation offers extensive *how-to* guides and technical information about the services provided by the Vlaams Supercomputer Centrum.

VSC Accounts

How to get your VSC account to use the different VSC services and platforms.

Compute

The high-performance computing (HPC) platform provides multiple tiers of parallel processing enabling researchers to run advanced application programs efficiently, reliably and quickly.

- Tier-1 HPC
- Tier-2 HPC
- Terminal Interface

Data

The VSC Data component enables research data to remain close to the computing infrastructure during the active phase of the data life cycle.

- Tier-1 Data

Cloud

The VSC Cloud component provides *on-demand* resources in a more flexible and cloud-like manner.

- Virtual Machines

Manage your VSC account at
<https://account.vscentrum.be>

The screenshot shows a web browser window with the URL account.vscentrum.be/django/ in the address bar. The page features the VLAAMS SUPERCOMPUTER CENTRUM logo and a navigation menu with the following items: View Account, Edit Account, View Groups, New/Join Group, Edit Group, New/Join VO, Reservations, and Log Out. Below the navigation menu, the 'View account' section is active, displaying 'General information' with fields for Uid, Institute (Leuven / Hasselt), Institute login, and Gecos. Several orange callout boxes with arrows point to specific elements: one points to the address bar, another to the 'View Account' menu item, a third to the 'Edit Account' menu item, a fourth to the 'New/Join Group' menu item, and a fifth to the 'Edit Group' menu item.

VSC View account — VSC

account.vscentrum.be/django/

VLAAMS SUPERCOMPUTER CENTRUM Vlaanderen is supercomputing

View Account Edit Account View Groups New/Join Group Edit Group New/Join VO Reservations Log Out

View account

General information

Uid:

Institute: Leuven / Hasselt

Institute login:

Gecos:

View public keys
Request account

Add/remove your
public keys

Create or
request to join
a new group

Moderate your VSC groups
(add/remove users)

Support and services

Basic support

- Helpdesk (hpcinfo@kuleuven.be)
- Monitoring and reporting

Application support

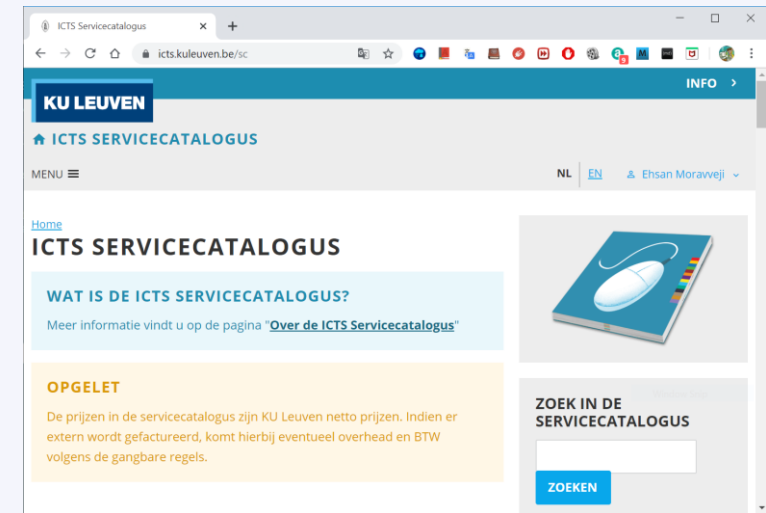
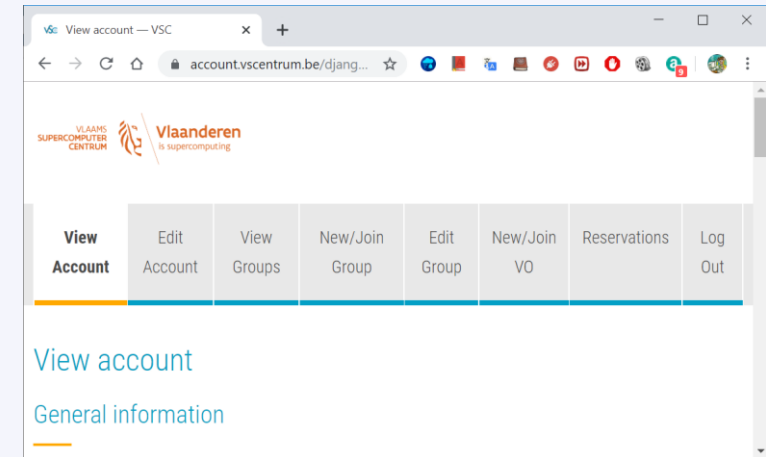
- Installation and porting
- Optimisation and debugging
- Benchmarking
- Workflows and best practices

Training

- Documentation and tutorials
- Scheduled trainings and workshops
- Workshops on request
- One-to-one sessions

Become a VSC user

- ☐ (Optional) Create a secure (4096 bit) [SSH key pairs](#)
Upload it on the account page: <https://account.vscentrum.be>
- ☐ You need to [request a VSC account](#)
Normally gets processed swiftly
- ☐ Request [introductory credits](#) (2M free credits for 6 months)
- ☐ Request [project credits](#) (for supervisors and project leaders)
You need to create a VSC group
Add users to the group to give them access to use credits
Fill out the request form
- ☐ Extra storage requests
Scratch extension: free of charge
Staging storage: 20 € per TB per year
- ☐ All service costs (compute and storage) are described
in the ICTS service catalog: <https://icts.kuleuven.be/sc>
Click on [High Performance Computing](#) (NL/EN)



VSC training

- Introductory

Matlab (Supercalculator)
Matlab programming

Linux

HPC intro

Linux for HPC

Make intro

CMake intro

Linux scripting

Linux tools

Version control systems

Infosessions:

- Containers
- Notebooks

- Intermediate

C++ for scientific computing

Fortran for programmers

C

- Python Introduction
- Python: System programming
- Scientific Python
- Python for software engineering
- Python for data science
- Python for machine learning

worker/atools

Julia good bad ugly

- Advanced

High Performance Python

- Specialized track

?

MPI

OpenMP

Debugging techniques

Code optimization

Stay up-to-date <https://www.vscentrum.be/en/education-and-trainings>

How to acknowledge VSC in publications

Why?

- ❑ a contractual obligation for the VSC
- ❑ helps VSC to secure funding
- ❑ you will benefit from it in the long run

For KU Leuven users

- ❑ add the relevant papers to the “KUL-HPC” virtual collection in Lirias

In het Nederlands

De infrastructuur en dienstverlening gebruikt in dit werk werd voorzien door het VSC (Vlaams Supercomputer Centrum), gefinancierd door het FWO en de Vlaamse overheid.

In English

The resources and services used in this work were provided by the VSC (Flemish Supercomputer Center), funded by the Research Foundation - Flanders (FWO) and the Flemish Government.



Tier-2 clusters

VSC HPC infrastructure

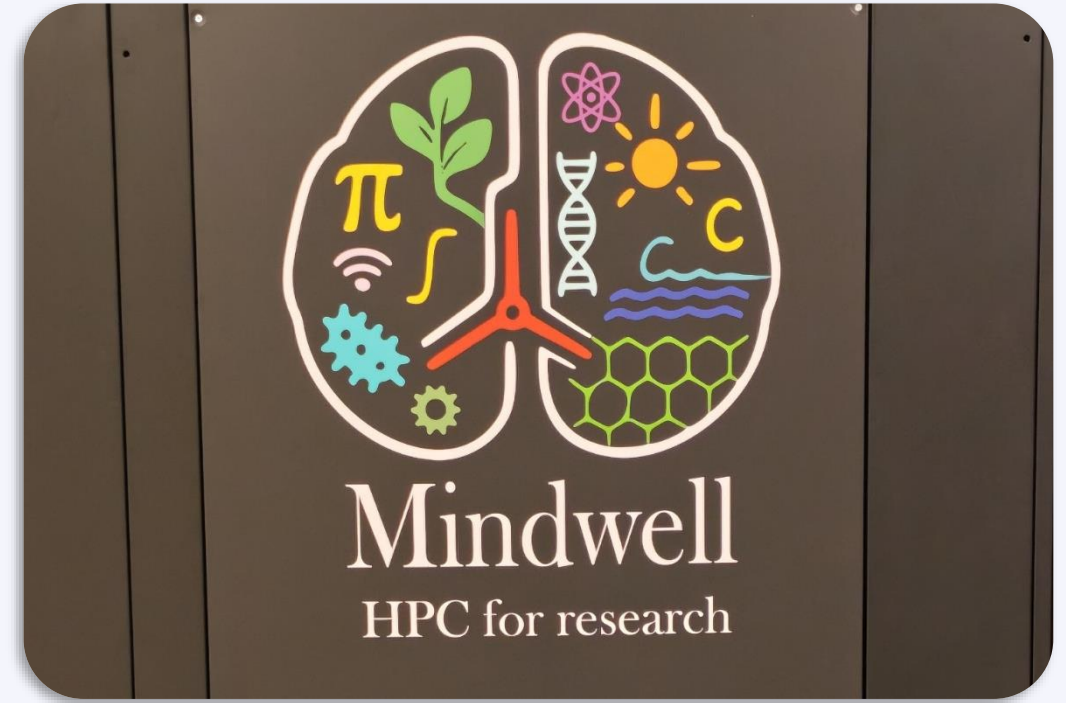


sof1a

Tier-2 clusters @ KU Leuven



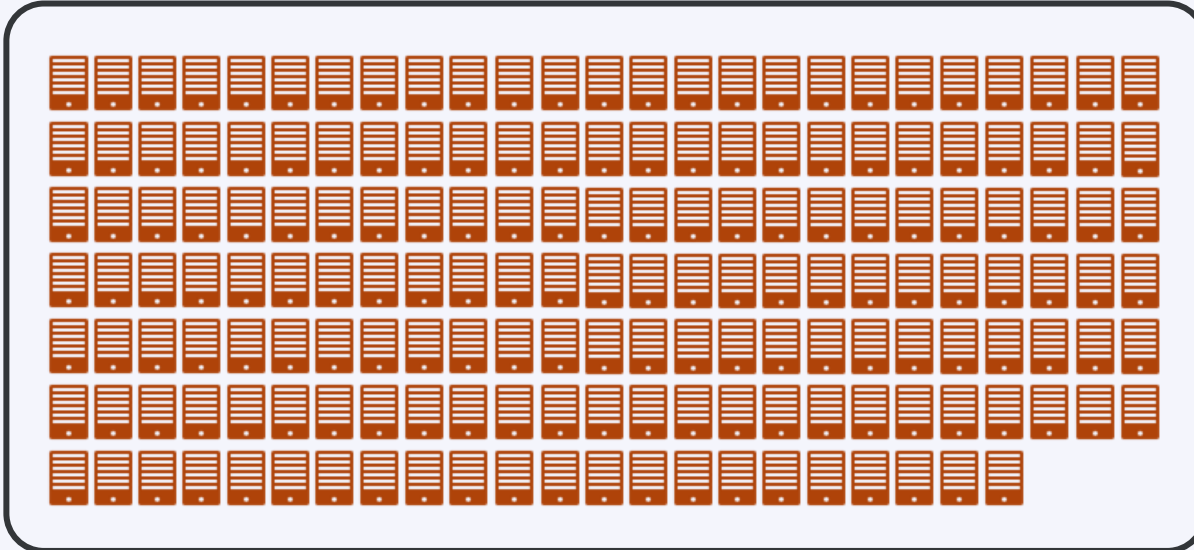
wICE (since September 2022)
276 nodes | 20848 cores



Mindwell (since May 2026)
55 nodes | 10560 cores

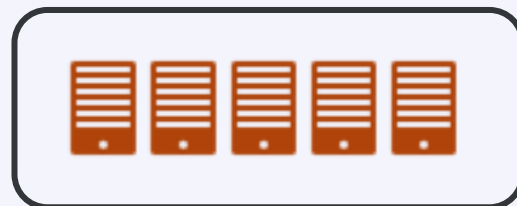
Tier-2 cluster – wICE CPU

Icelake Thin nodes



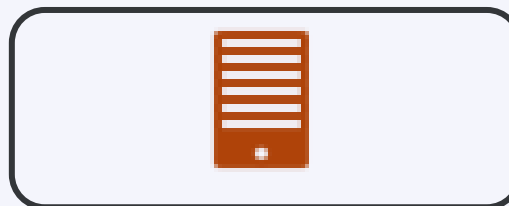
172x IceLake 72c 256 GiB

Large memory nodes



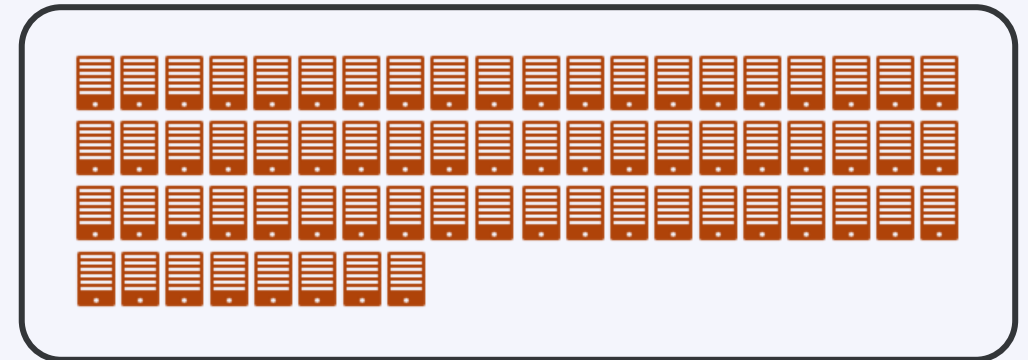
5x IceLake 72c 2 TiB

Huge memory node



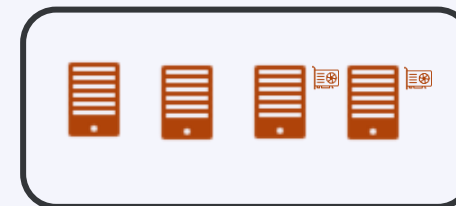
1x IceLake 72c 8 TiB

Sapphire Rapids Thin nodes



68x Sapphire Rapids 96c 256 GiB

Login nodes

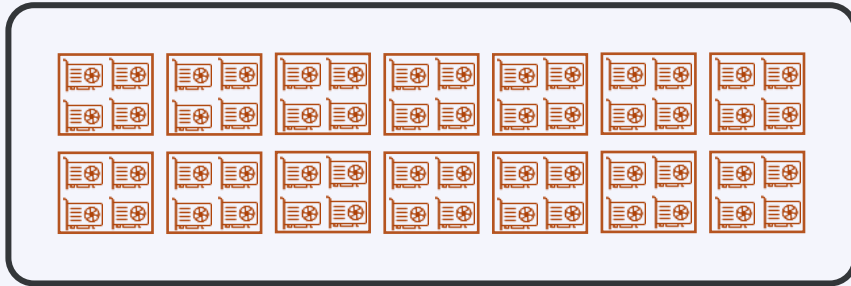


2x SkyLake 36c 377 GiB

2x SkyLake 36c 377 GiB + P6000 24 GiB

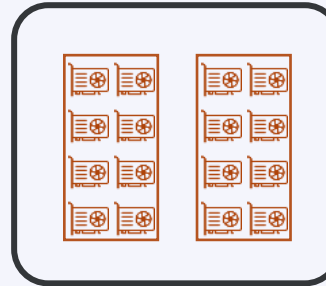
Tier-2 cluster – wICE GPU

P100 GPU nodes



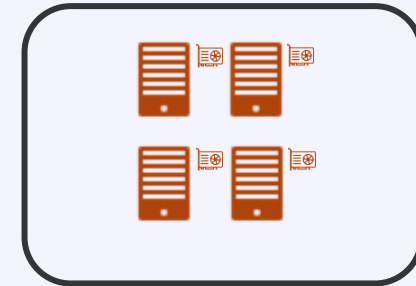
14x SkyLake 36c 256 GiB
4 P100 SXM2 16 GiB

V100 GPU nodes



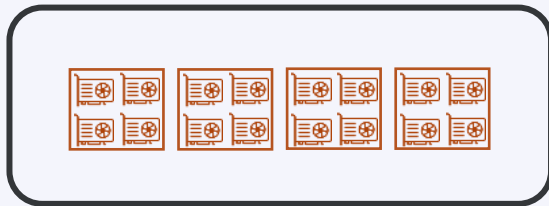
2x CascadeLake 36c 256 GiB
8 V100 SXM2 32 GiB

Interactive nodes



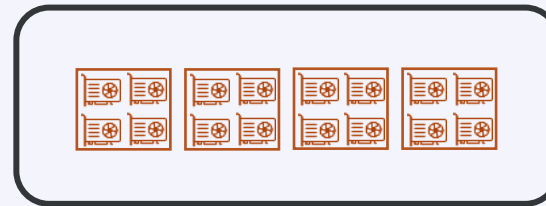
4x IceLake 64c 512 GiB
1 A100 80GB

A100 GPU nodes



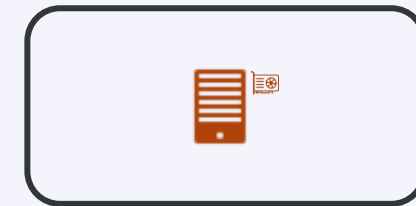
4x IceLake 72c 512 GiB
4 A100 SXM4 80 GiB

H100 GPU nodes



4x AMD Genoa 64c 768 GiB
4 H100 SXM5 80 GiB

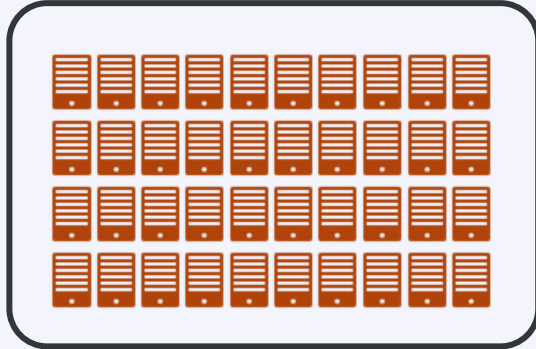
A100 Debug node



1x IceLake 64c 512 GiB
1 A100 80GB

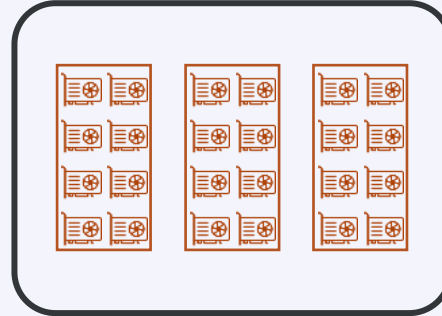
Tier-2 cluster – Mindwell CPU

Granite Rapids Thin nodes



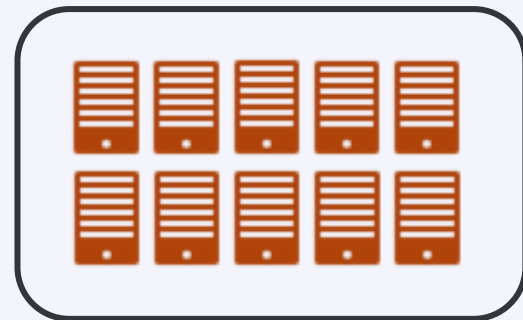
40x Granite Rapids 192c 768 GiB

B200 GPU nodes



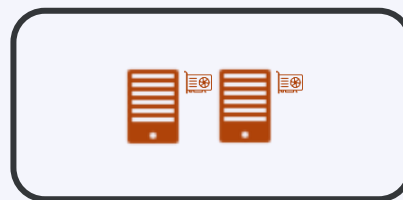
3x AMD Turin 192c 1.5 TiB
8 B200 SXM6 192 GiB

Large memory nodes



10x Granite Rapids 192c 1.5 TiB

Interactive nodes



2x Granite Rapids 192c 768 GiB
2 RTX5000 32 GiB

Login nodes



Tier-2 cluster – wICE CPU

Node type	CPU type	# cores	Total mem	Mem per core (MiB)	Partition
Icelake	Xeon 8360Y	72	256 GiB	3400	batch_icelake
Sapphire Rapids	Xeon 8468	96	256 GiB	2500	batch_sapphirerapids
Icelake large mem	Xeon 8360Y	72	2 TiB	28000	bigmem
Icelake huge mem	Xeon 8360Y	72	8 TiB	111900	hugemem

- **OS:** Rocky 9.6
- **Access:** login.hpc.kuleuven.be
- **Interconnect:** InfiniBand HDR (100 Gb/s)
- **SSD Disks:** 960 GB

Remarks

Tier-2 cluster – wICE GPU

Node type	CPU & GPU type	# cores	Total mem	Mem per core (MiB)	Partition
Skylake P100 GPU	Xeon Gold 6140 4xP100 SXM2 16GB	36	192 GiB	5000	gpu_p100
Cascadelake V100 GPU	Xeon Gold 6240 8xV100 SXM2 32GB	36	768 GiB	21000	gpu_v100
Icelake A100 GPU	Xeon 8360Y 4xA100 SXM4 80GB	72	512 GiB	7000	gpu_a100
Genoa H100 GPU	Epyc 9334 4xH100 SXM5 80GB	64	768 GiB	11700	gpu_h100
Icelake interactive	Xeon 8358 1xA100 SXM4 80GB	64	512 GiB	2000	Interactive
Icelake A100 debug	Xeon 8358 1xA100 SXM4 80GB	64	512 GiB	7500	gpu_a100_debug

- **OS:** Rocky 9.6
- **Access:** login.hpc.kuleuven.be
- **Interconnect:** InfiniBand HDR (100 Gb/s)

- **SSD Disks:** 960 GB
- **Interactive partition:**
max resources: 8 cores, 1 GPU, 16hr

Remarks

Tier-2 cluster – Mindwell

Node type	CPU type	# cores	Total mem	Mem per core (MiB)	Partition
Granite Rapids	Xeon 6972P	192	768 GiB	4000	batch_graniterapids
Icelake large mem	Xeon 6972P	192	1.5 TiB	8000	bigmem
Icelake A100 GPU	EPYC 9655 8xB200 SXM6 192GiB	192	1.5 TiB	8000	gpu_b100
Icelake interactive	Xeon 6972P 2x RTX5000 32GiB	192	768 GiB	4000	Interactive

- **OS:** Rocky 9.6
- **Access:** login.hpc.kuleuven.be
- **Interconnect:** InfiniBand NDR (400 Gb/s)

- **SSD Disks:** 960 GB
- **Interactive partition:**
max resources: 8 cores, 1 GPU, 16hr

Remarks



Storage

Overview of the Tier-2 storage locations

- ✓ Only you own your files (POSIX)
Users can share folders via [VSC groups](#)
- ✓ A VSC account has 3 default storages (free of charge)
 - \$VSC_HOME
 - \$VSC_DATA
 - \$VSC_SCRATCH
- ✓ You can additionally request `staging` storage
- ✓ Different storage volumes have different:
 - mount point
 - size and performance
 - use case
 - backup and maintenance policy
- ✓ For more info, see the [ICTS Service Catalog](#) (EN/NL)

Storage locations

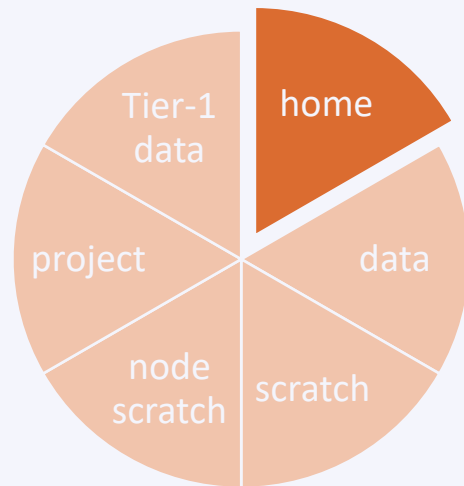
- ❑ **Avoid using `/tmp` (`$TMPDIR`) on the login nodes**
It is only 10 GB and is needed by the operating system
Your application can crash if using `/tmp`
- ❑ You are automatically logged into your home folder upon login. Best to immediately switch to some other storage, e.g.
`$ cd $VSC_DATA`
- ❑ Always check your storage balance using `myquota` command

Example

```
$ myquota
file system $VSC_HOME
    Blocks: 1479M of 3072M
    Files: 12934 of 100k
file system $VSC_DATA
    Blocks: 12G of 75G
    Files: 1043k of 10000k
file system $VSC_SCRATCH
    Blocks: 15M of 1.5T
```

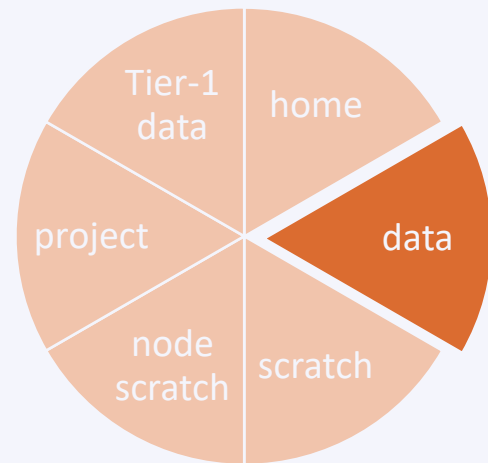
- ✓ [Request form for extra storage](#)
- ✓ [More info in the service catalog](#)

Storage locations



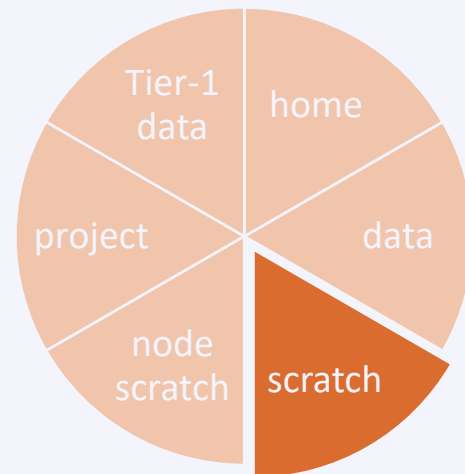
Storage	Home folder
Env. variable	<code>\$VSC_HOME</code>
Filesystem type	NFS
Access	Global
Backup	Hourly, daily, weekly (until last month) inside <code>.snapshot</code> folders
Default quotum	3 GB
Quotum increase	Not possible
Usage	Only storing SSH keys, config files, ...
Remarks	- Stay away from using it - Can easily overflow: + Your jobs may crash + Login issues

Storage locations



Storage	Data folder
Env. variable	\$VSC_DATA
Filesystem type	NFS
Access	Global
Backup	Hourly, daily, weekly (until last month) inside <code>.snapshot</code> folders
Default quorum	75 GB
Quorum increase	On purchase
Usage	Your data, code, software, results
Remarks	- Permanent storage for initial/final results - Not optimal for intensive or parallel I/O

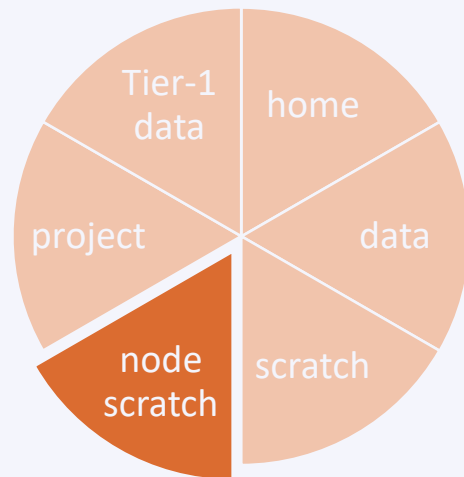
Storage locations



The Lustre filesystem is nearly full.
Momentarily, we don't extend user quota's
with more than 1TB.

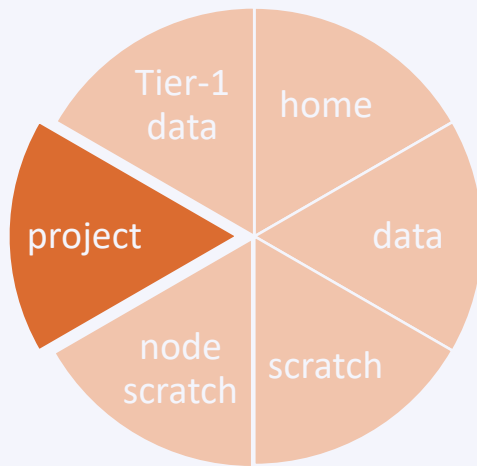
Storage	Scratch folder
Env. variable	\$VSC_SCRATCH \$VSC_SCRATCH_LUSTRE1 \$VSC_SCRATCH_GPFS1
Filesystem type	Lustre & GPFS
Access	Local
Backup	Files get deleted 30 days after last access
Default quorum	500 GB for both
Quotum increase	Free
Usage	Intensive, parallel I/O, temporary files
Remarks	- Recommended storage for all jobs - Copy scratch files to VSC_DATA or local storage after jobs are done - Deleted files cannot be recovered - Avoid folders with >10,000 files

Storage locations



Storage	Node scratch folder
Env. variable	<code>\$VSC_SCRATCH_NODE</code>
Filesystem type	Ext4
Access	On compute node, only at runtime
Backup	None
Default quorum	591 GB
Extension	Read about beeOND
Usage	Temporary storage at runtime
Remarks	- Fastest I/O, attached to the node - Is cleaned after job terminates - Copy the data to your home, scratch, or staging before job ends

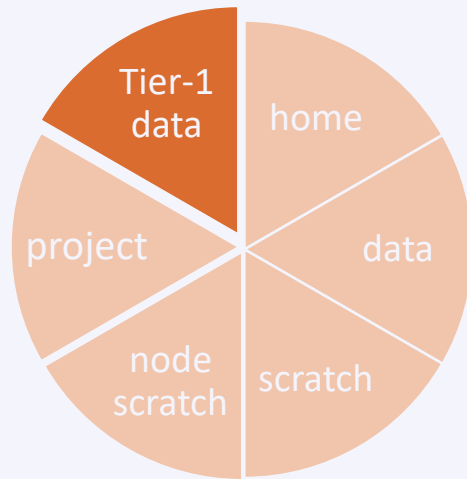
Storage locations



The Lustre filesystem is nearly full.
Momentarily, we don't create or extend
users' staging quota requests.

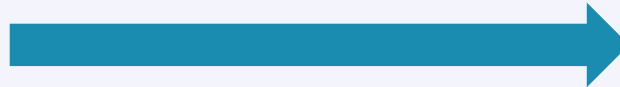
Storage	project folder
Path	/lustre1/project/stg_000XX /gpfs1/project/stg_000XX
Filesystem type	Lustre & GPFS
Access	On demand, only Tier-2@KUL
Backup	None
Default quorum	None
Extension	On purchase, from 1 TB
Usage	Permanent; share with a group
Remarks	- Accessible from login/compute nodes - Fast, parallel I/O

Storage locations



- ❑ VSC offers [Tier-1 Data](#) as a large storage service
- ❑ Only for active data (no archiving)
- ❑ Supports metadata (search & discover)
- ❑ Share within VSC (permission management)
- ❑ Share outside VSC (via [Globus](#))
- ❑ Stage-in/stage-out data workflow
- ❑ Free of charge for academic use
- ❑ Starting volume is 5 TB
- ❑ Submit a proposal to get access
You can apply at any time

Extra (KU Leuven) storage locations



- In production
- For **active** data
- Data should stage in/out
- Has backup
- See the [ICTS service catalog](#)

- In development
- For **inactive** data
- Storage until 10 years
- Has backup
- See the [ICTS service catalog](#)

Also see the [KU Leuven storage guide](#) for even more storage possibilities

ManGO





Compute credits

Tier-2 credit system

What?

- Credits are a measure for compute resources
- Mandatory for accessing these resources
- Traceability and accountability

Why?

- Incentive to optimize your code and workflow

How?

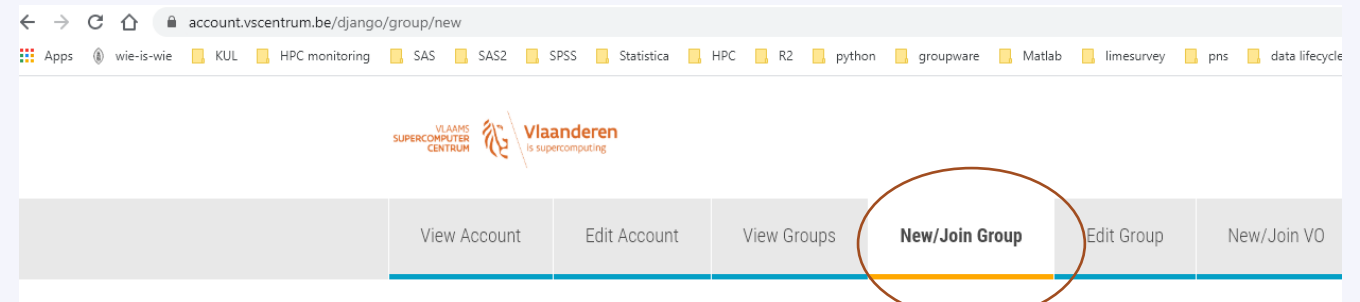
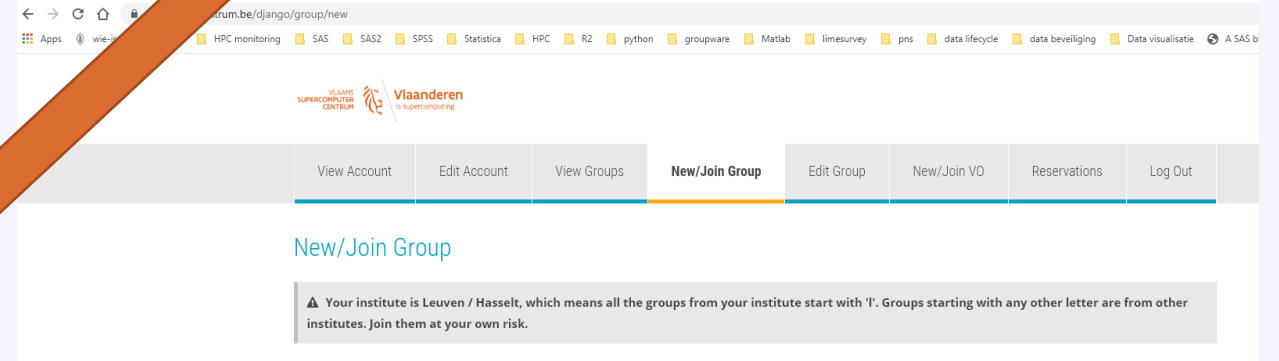
- 1 million credits = 3.5 EUR
- Pay-as-you-go model
- Credit rates on the right = for fully allocated nodes (if you request less cores, you also get billed less)
- See the [ICTS service catalog](#) for more info

Node types	Credits per node-hour
wICE Skylake 4 GPUs	20 000
wICE Cascadelake 8 GPUs	39 900
wICE IceLake thin node	11 000
wICE IceLake Large Memory	19 000
wICE IceLake GPU node	45 000
wICE IceLake interactive node	Free
wICE IceLake hugemem node	80 000
wICE Sapphire Rapids	20 000
wICE AMD Zen-4 4 GPUs	150 000

Tier-2 credit system – how to request credits

- Request your introduction credits
 - Every new user can request them once
 - Free of charge
 - 6 months
- Request permission to join a group
- Request project credits
 - Create group on the VSC account page
 - Start the name preferably with **p_**
 - Your group will start with **lp_**
 - Creator of the group can add and remove users
- Tier-2 credits can be used on all (Leuven) Tier-2 clusters

Check the [ICTS service catalog](#)



Create new group

⚠ We will automatically prepend the letter 'I' to your groupname.

Groupname *

Tier-2 credit system – how

- Submit your jobs with a suitable credit account

```
#SBATCH -A lp_myproject
```

```
sbatch -A lp_myproject ... myjob.slurm
```

- Before submitting the job, Slurm checks if the account has enough available credits for the job (in addition to any other already-running jobs)
- Once a job start running, it starts consuming credits from the account

Tier-2 credit system – manage your credits

- We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-balance [-A account]</code>	List active projects and available credits

```
tier2-p-login-3$ sam-balance -A lp_hpcinfo
ID          Name          Balance      Reserved      Available
=====
96141      lp_hpcinfo      9782636      0             9782636
tier2-p-login-3$
```

Tier-2 credit system – manage your credits

- We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-list-allocations -A <account-name></code>	List the credit deposits and refunds (if any)

```
tier2-p-login-3$ sam-list-allocations -A lp_hpcinfo
AllocID   AccountID Account          Timestamp                Credits
=====
80745     96141      lp_hpcinfo      2023-02-14T10:40:50      2000000
203565    96141      lp_hpcinfo      2023-04-17T16:42:26      2000000
227730    96141      lp_hpcinfo      2023-05-03T09:10:29      2000000
313474    96141      lp_hpcinfo      2023-06-19T17:33:59      8410000
tier2-p-login-3$
```

Tier-2 credit system – manage your credits

- We provide various *sam*-commands to facilitate credit related queries

Command	Purpose
<code>sam-quote sbatch [sbatch args] [myjob.slurm]</code>	Estimates the (max) amount of credits a job will use

```
tier2-p-login-3$ sam-quote sbatch --cluster=wice hello.slurm
10980
tier2-p-login-3$
```

Tier-2 credit system – manage your credits

- We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-statement -A <account-name></code> <code>-s <start-date></code> <code>-e <end-date></code>	Overview of credits used by each job in some time window

```
tier2-p-login-3$ sam-statement -A lp_hpcinfo -s 2023-09-01 -e 2023-10-25
#####
#
# Includes Account=lp_hpcinfo
# Generated on Mon Oct 30 11:01:29 2023.
# Reporting fund activity from 2023-09-01 to 2023-10-25
#
#####

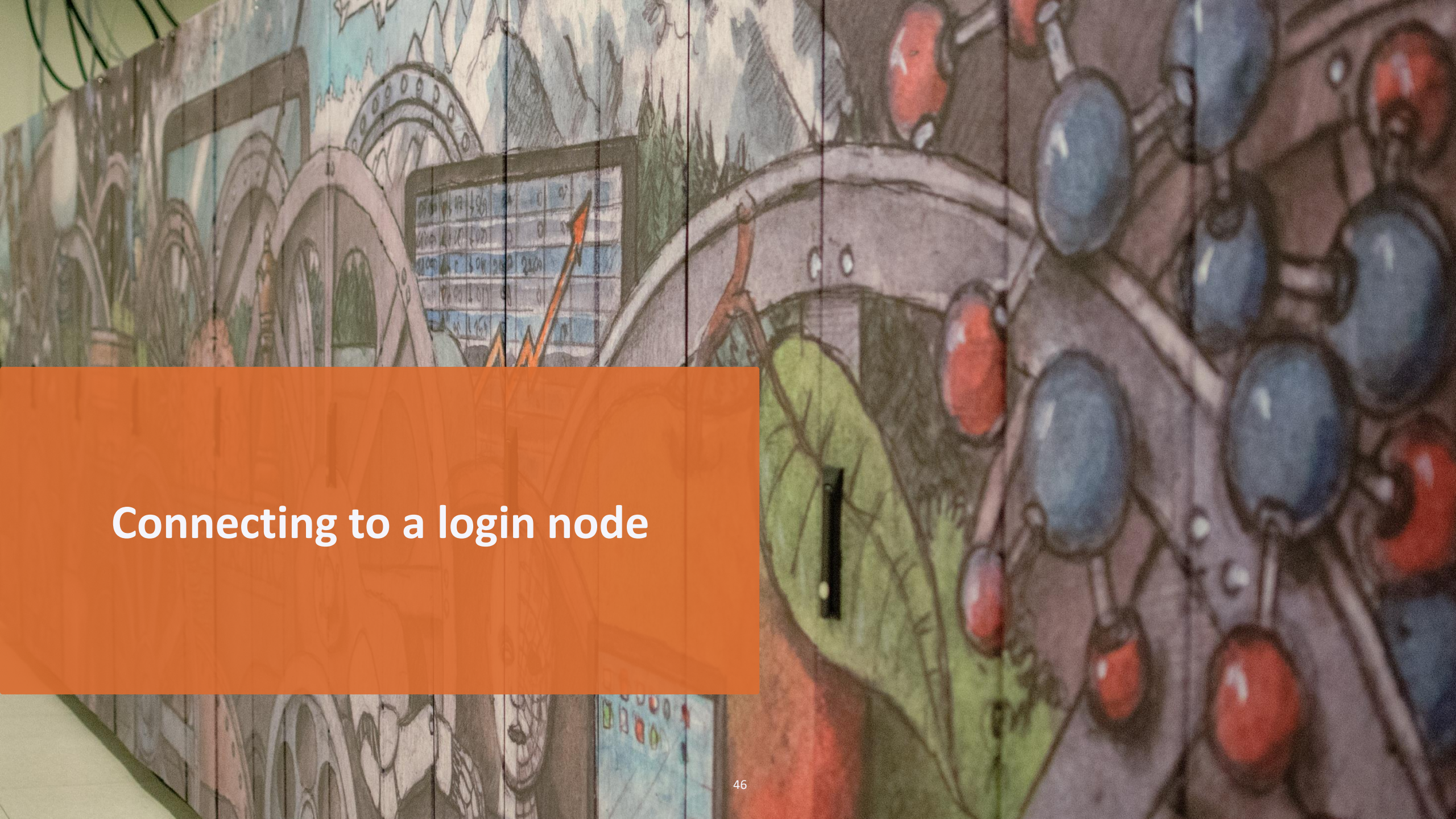
-----
Credits deposited in the given period:          0
Credits refunded in the given period:           0
Credits consumed in the given period:           0
-----

JobID      Cluster  Account          User      Partition  Credits
=====
60739746   wice      lp_hpcinfo       vsc30446  batch      0
60740123   wice      lp_hpcinfo       vsc30446  batch      0
60740125   wice      lp_hpcinfo       vsc30446  batch      0
60746730   wice      lp_hpcinfo       vsc30446  interactive 0
60773547   wice      lp_hpcinfo       vsc30446  interactive 0
60774141   wice      lp_hpcinfo       vsc30446  batch      0
60774160   wice      lp_hpcinfo       vsc30446  batch      0
60783857   wice      lp_hpcinfo       vsc30446  interactive 0
```

Tier-2 credit system – manage your credits

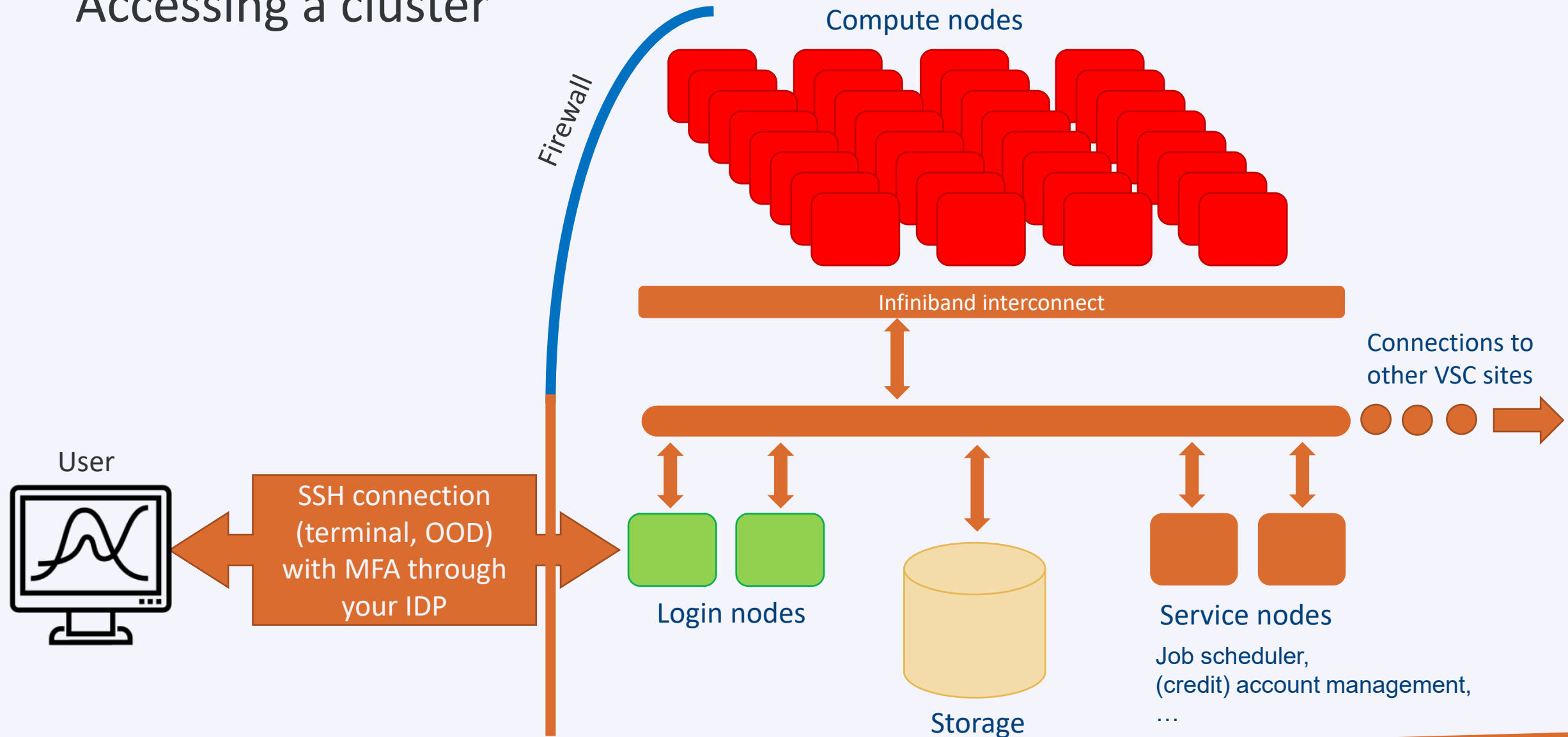
- We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-balance</code>	Lists active projects and available credits
<code>sam-list-allocations -A <account-name></code>	Lists the credit deposits and refunds (if any)
<code>sam-quote sbatch [sbatch args] [myjob.slurm]</code>	Estimates the (max) number of credits a job will use
<code>sam-statement -A <account-name> -s <start-date> -e <end-date></code>	Overview of credits used by each job in some time window



Connecting to a login node

Accessing a cluster



How to login?

Open OnDemand

- ☐ Web-based login via your browser
- ☐ <https://ondemand.hpc.kuleuven.be>
- ☐ Multi-Factor Authentication (MFA):
Login through your institute's IDP
- ☐ No SSH key needed

SSH client

- ☐ Windows: [PuTTY](#), [MobaXterm](#), ...
MacOS/Linux: any terminal
- ☐ Optional: for VSC clusters *not* hosted at KU Leuven you need an [\(open\)SSH key pair](#) (upload the public key to the VSC account page and wait ≥ 30 mins)
- ☐ Hosts: login.hpc.kuleuven.be:22
Recommended to use an [SSH agent](#)
- ☐ Can open GUIs if X11 forwarding is enabled

Using the login nodes

- ☐ To develop code
- ☐ To check your storage and credit balance
- ☐ To manage jobs (submit, check status, debug, resubmit, ...)
- ☐ To move data around
 - within VSC: use data, scratch, staging
 - outside VSC: copy/sync from/to your local storage (e.g. Globus)
- ☐ To pre- or postprocess your data
- ☐ To visualize your data
- ☐ To share files and folders

Tips

Do NOT execute CPU- or memory-intensive tasks here

Warning

Login nodes are shared resources

Submit jobs instead (you can e.g. compile software on compute nodes via an interactive job)

Use `slurmtop` to see how busy the cluster is

(Linux and Mac)

- ✓ Use ssh to connect:
`$ ssh -A vscXXXXXX@<host_name>`
- ✓ If key not found:
`$ ssh -i </path/to/keyfile> ...`

If asked for **password**, please stop connecting and contact support, otherwise after a few attempts you will be blocked for 24h.

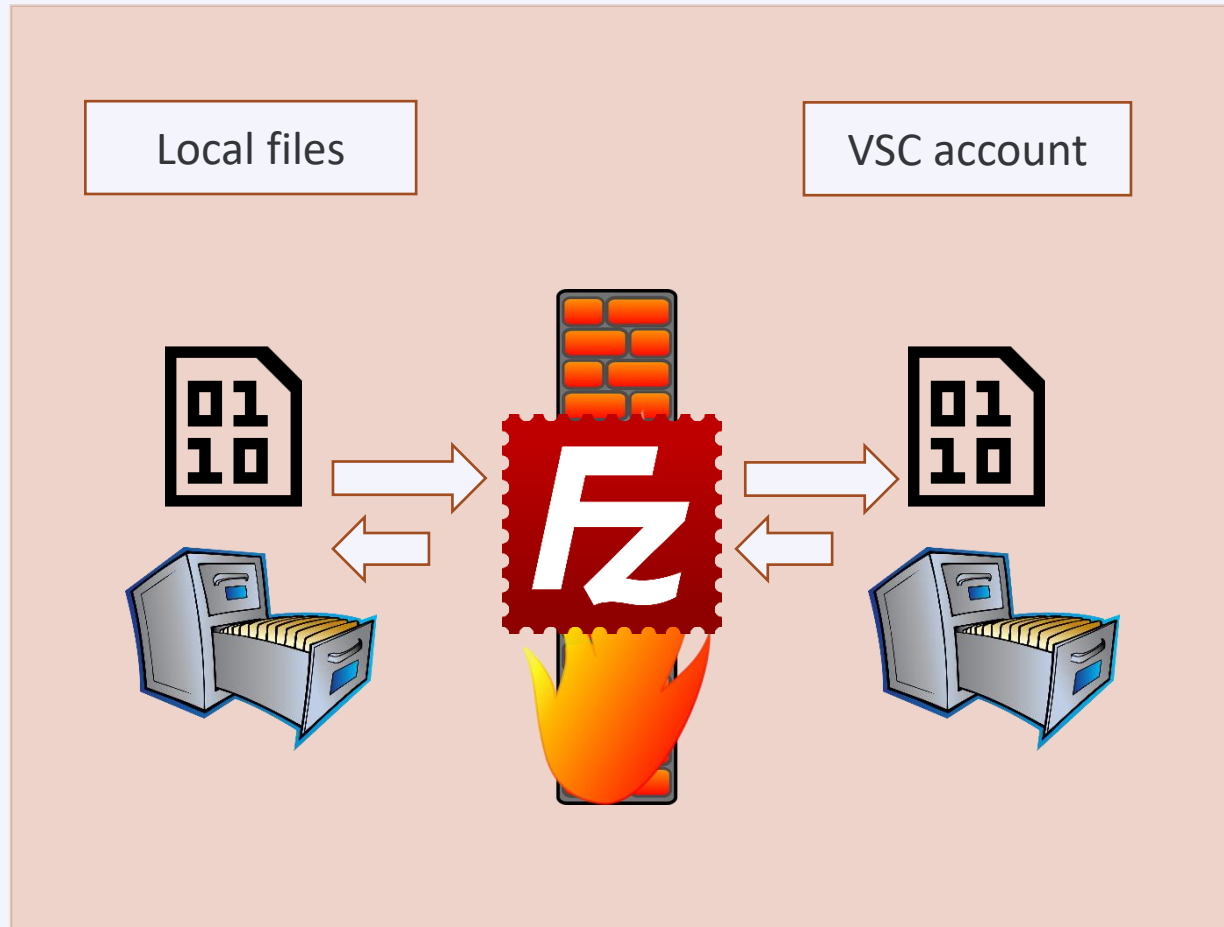
Hostname:
login.hpc.kuleuven.be

[illegible]



Data transfer

File transfer tools

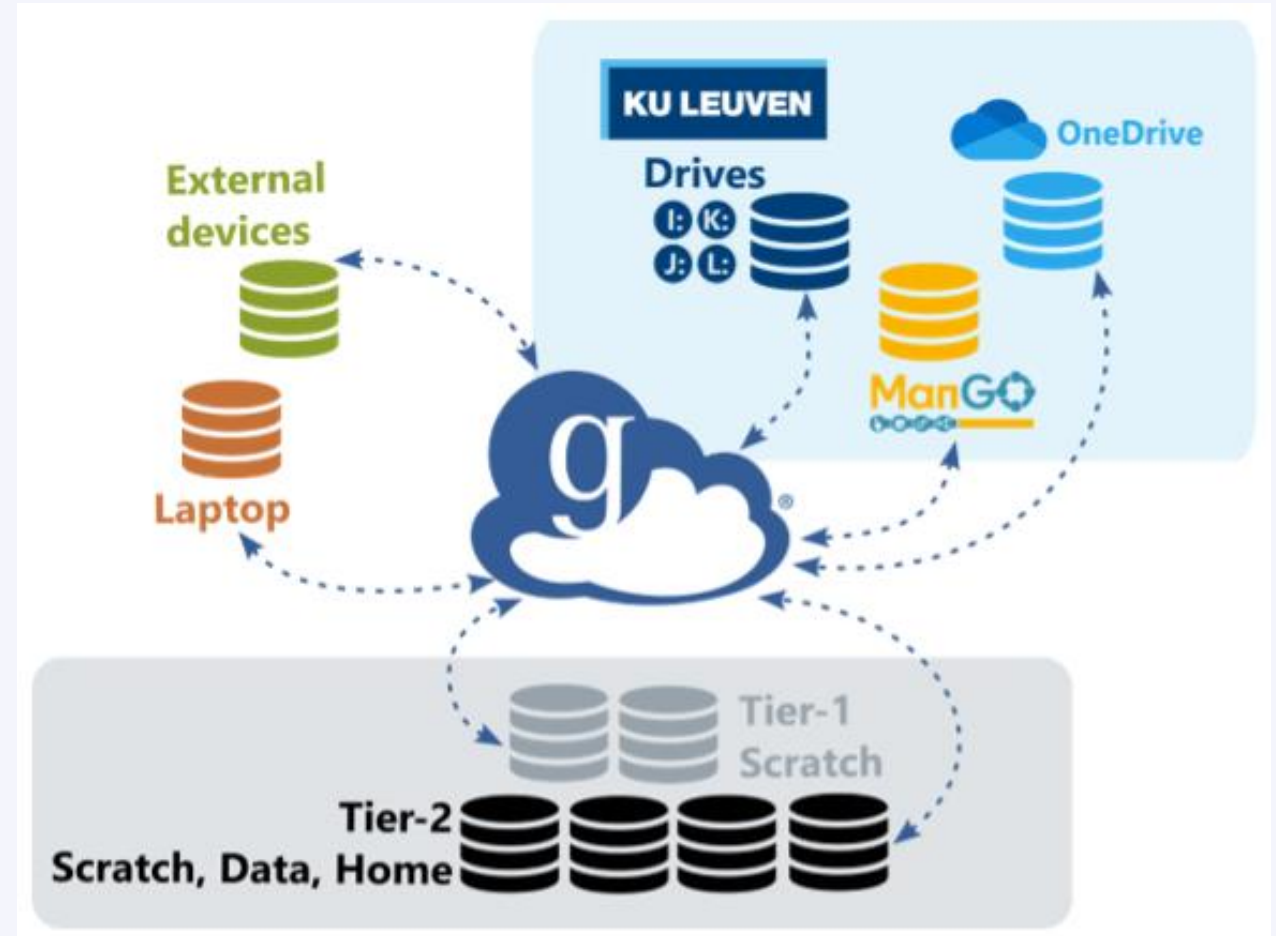


Application	OS
FileZilla	Windows, Linux, Mac
WinSCP	Windows
rsync, scp	Linux, Mac
Globus	Window, Linux, Mac

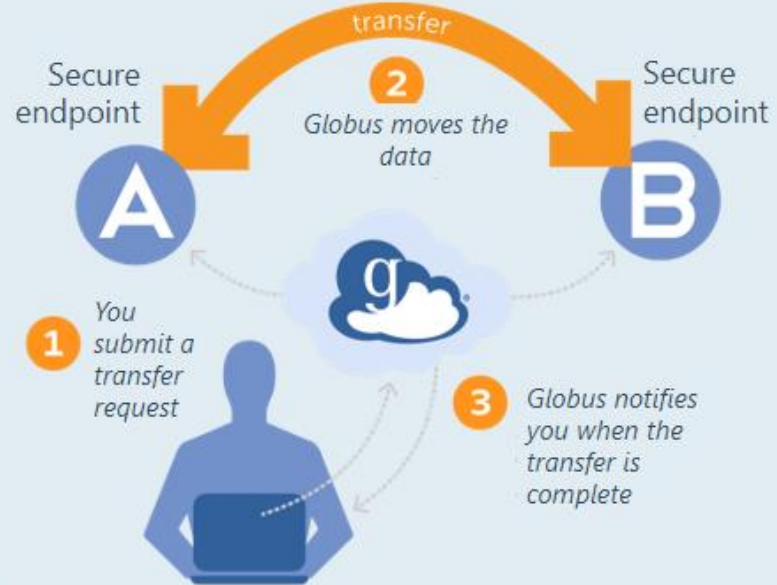
File transfer with Globus

- ❑ Web interface for transferring, sharing and searching within data
- ❑ Schedule and resume (large) transfers
- ❑ Workflow:
 - Define “endpoint” on the source
 - Define “endpoint” on the destination
 - Transfer between endpoints
- ❑ Endpoints on all VSC sites:
 - all Tier-2 clusters: home, data, scratch*
 - all Tier-1 clusters: scratch, projects

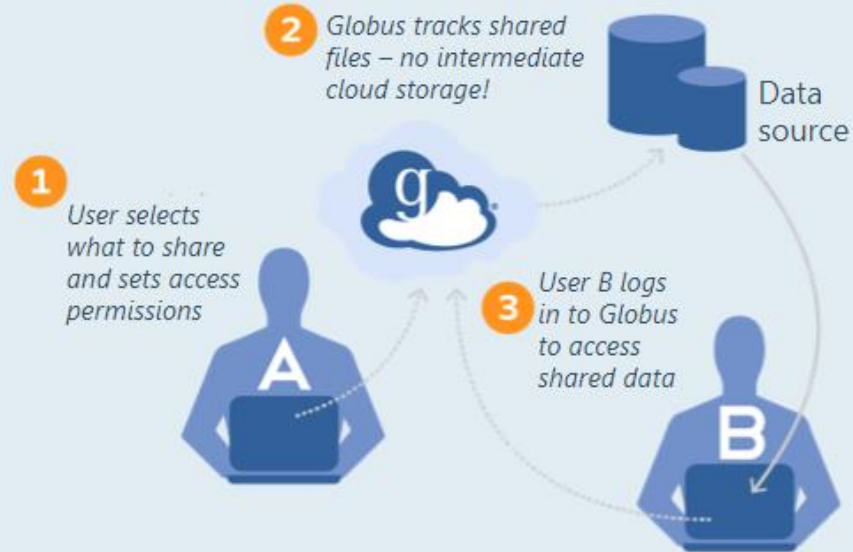
* You can also access staging storage through this endpoint (Leuven Tier-2 only)



Transfer data



Share data





Open OnDemand

Open OnDemand

- ✓ Access clusters via your web browser
- ✓ <https://ondemand.hpc.kuleuven.be>
- ✓ Login via your institute's IDP
- ✓ File browser + Data Transfer (Globus)
- ✓ Develop, compile and test your code
- ✓ Interactive apps + integrated shell
- ✓ noVNC Desktop for graphical apps
- ✓ VSCode and PyCharm editors
- ✓ Fluent, Jupyter Lab, Rstudio, MATLAB, ParaView, Tensorboard, Gnuplot

Please close your interactive sessions
when you no longer actively use them!



KU LEUVEN

Open OnDemand provides an integrated, single access point for all your HPC resources.

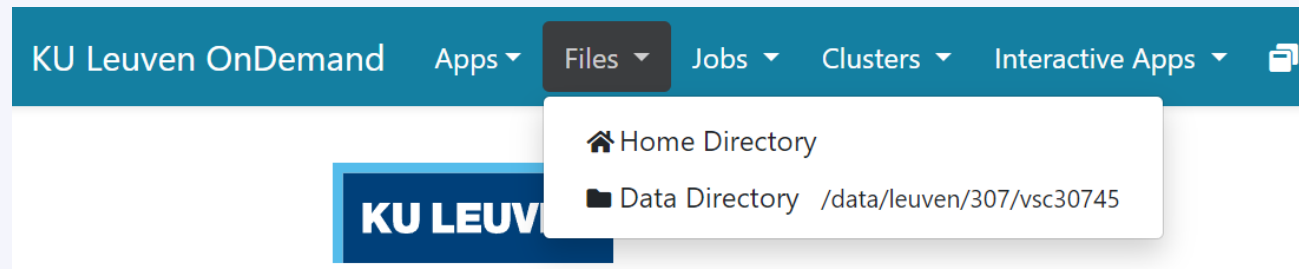
Pinned Apps A featured subset of [all available apps](#)

 Active Jobs System Installed App	 Home Directory System Installed App	 Login Server Shell Access System Installed App	 MATLAB System Installed App
 Fluent System Installed App	 Visual Studio Code System Installed App	 Desktop System Installed App	 gnuplot System Installed App
 IGV-Web System Installed App	 Interactive Shell System Installed App	 JupyterLab System Installed App	 ParaView System Installed App
 PyCharm System Installed App	 RStudio Server System Installed App	 TensorBoard System Installed App	 VS Code Tunnel System Installed App

Open OnDemand

Access your VSC storage

- ✓ Use the “Files” tab to access your VSC_HOME and VSC_DATA
- ✓ VSC_SCRATCH is not available from this menu



File/Folder management



- ✓ OK for transferring small files/folders
- ✓ Open any (sub)folder in terminal (will open a new tab)
- ✓ Deleted files/folders can be retrieved from (hourly, daily, weekly) snapshots

Monitor your active jobs

KU Leuven OnDemand Apps ▾ Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾   

⌚ Active Jobs
🔧 Project Manager

Your Jobs ▾ All Clusters ▾

Filter:

Active Jobs

Show entries

ID	Name	User	Account	Time Used	Queue	Status	Cluster	Actions
 62138076	sys/dashboard/sys/rstudio	vsc32591	lp_hpcinfo	00:00:04	interactive	Running	wice	

Showing 1 to 1 of 1 entries

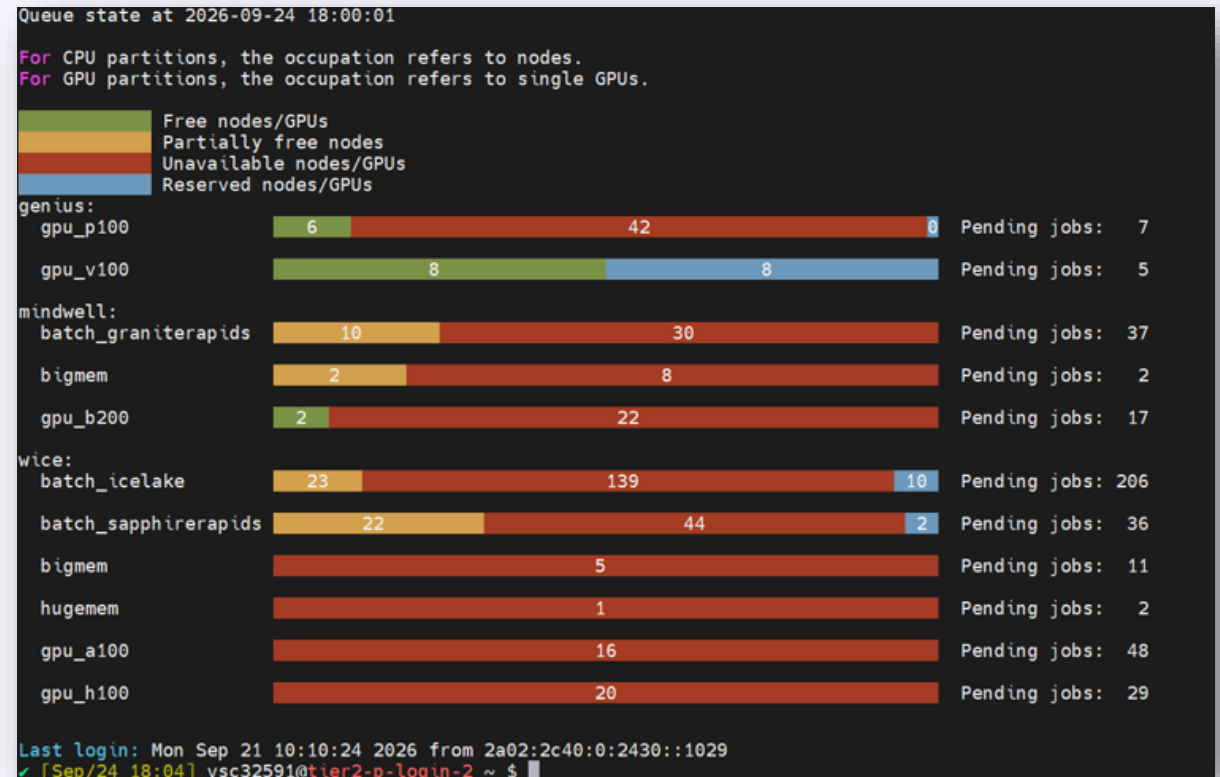
Previous 1 Next



Open OnDemand: Login Server Shell Access

Login shell

- ✓ Starts a shell in a new browser tab
- ✓ You land on a wICE login node (from which you can also access Mindwell)
- ✓ You land on your `VSC_HOME`.
`cd` to `data/scratch/staging` immediately
- ✓ Do NOT compute on the login nodes
- ✓ Try:
`$ sam-balance`
`$ myquota`





Open OnDemand: Interactive Shell

Interactive Shell

- ✓ Start a shell as a job (on a compute node)
- ✓ Recommended to:
 - Install your own software (in `VSC_DATA`)
 - Test/run your program interactively
 - Debug a program
- ✓ Choose the relevant cluster, partition and resources (cores, memory, GPU)

Interactive Apps

Servers

Interactive Shell

Jupyter Lab

Nvidia Rapids

RStudio Server

Tensorboard

code-server

Interactive Shell

This app will launch an interactive shell on a node.

Cluster

vice

Account

lpt2_sysadmin

Partition

interactive

batch(_long) or bigmem or interactive or gpu(_p100[_v100]) or dedicated...

Number of hours

1

Number of cores

1

Required memory per core in megabytes

3400

Number of nodes

1

Number of gpu's

0

[type:] Specify the total number of GPUs slices for the job. An optional GPU type specification can be supplied. For example "A100:3" or "3".

Reservation (optional)

Name of an existing reservation in which the job should run

☐ I would like to receive an email when the session starts

Launch

* The Interactive Shell session data for this session can be accessed under the [data root directory](#).

Open OnDemand: JupyterLab

Jupyter Lab

- ✓ Create a notebook as a job (on a compute node)
- ✓ Suitable for:
 - pre/post-processing your data
 - prototyping
 - adding figures and text for pedagogical purposes
- ✓ Choose the appropriate cluster, partition and resources (cores, memory, GPU)
- ✓ User your own Python/R kernels (next slide)

Interactive Apps

Servers

- Interactive Shell
- Jupyter Lab**
- Nvidia Rapids
- RStudio Server
- Tensorboard
- code-server

Jupyter Lab

This app will launch a Jupyter Lab server on one or more nodes.

Cluster

genius

Account

lpt2_sysadmin

Partition

gpu_p100

batch[_long] or bigmem or interactive or gpu[_p100[_v100]] or dedicated...

☐ Enable nvidia rapids

Only available on wICE interactive or gpu nodes with a GPU requested

Number of hours

1

Number of cores

9

Required memory per core in megabytes

3400

Number of nodes

1

Number of gpu's

1

[type:] Specify the total number of GPUs slices for the job. An optional GPU type specification can be supplied. For example "A100:3" or "3".

Reservation (optional)

Name of an existing reservation in which the job should run

☐ I would like to receive an email when the session starts

Pre-run scriptlet

Bash commands to be executed before starting application, \$node_arch variable containing for example 'skylake' is available

Launch

* The Jupyter Lab session data for this session can be accessed under the [data root directory](#).



Open OnDemand: Custom R/Python kernels

Custom kernels

- ✓ The default kernels may not offer the R/Python/... interpreter version or packages you want
- ✓ You can create a miniconda env and add it to your OnDemand kernels

Step 1: Miniconda

Start an Interactive Shell

Go to your data folder

```
$ cd ${VSC_DATA}
```

Download Miniconda

```
$ wget https://repo.continuum.io/miniconda/Miniconda3-latest-Linux-x86\_64.sh
```

Install Miniconda in your VSC_DATA

```
$ bash Miniconda3-latest-Linux-x86_64.sh -b -p ${VSC_DATA}/miniconda3
```

Permanently add the path to Miniconda to your ~/.bashrc

```
$ echo 'export PATH="${VSC_DATA}/miniconda3/bin:${PATH}"' >> ~/.bashrc
```



Open OnDemand: Custom R/Python Kernels

Step 2: Environment setup

Create a new environment `<env_name>` with all packages you need, e.g.

```
$ conda create --name=<env_name> numpy scipy ...
```

Enable your new environment

```
$ source activate <env_name>
```

Add the ipykernel package

```
$ conda install -c conda-forge ipykernel
```

Step 3: Register your kernel

```
$ python -m ipykernel install --user --env PYTHONPATH "" \  
    --name <env_name> --display-name <env_name>
```

Step 4: Launching JupyterLab

Now start a new Jupyter Lab session

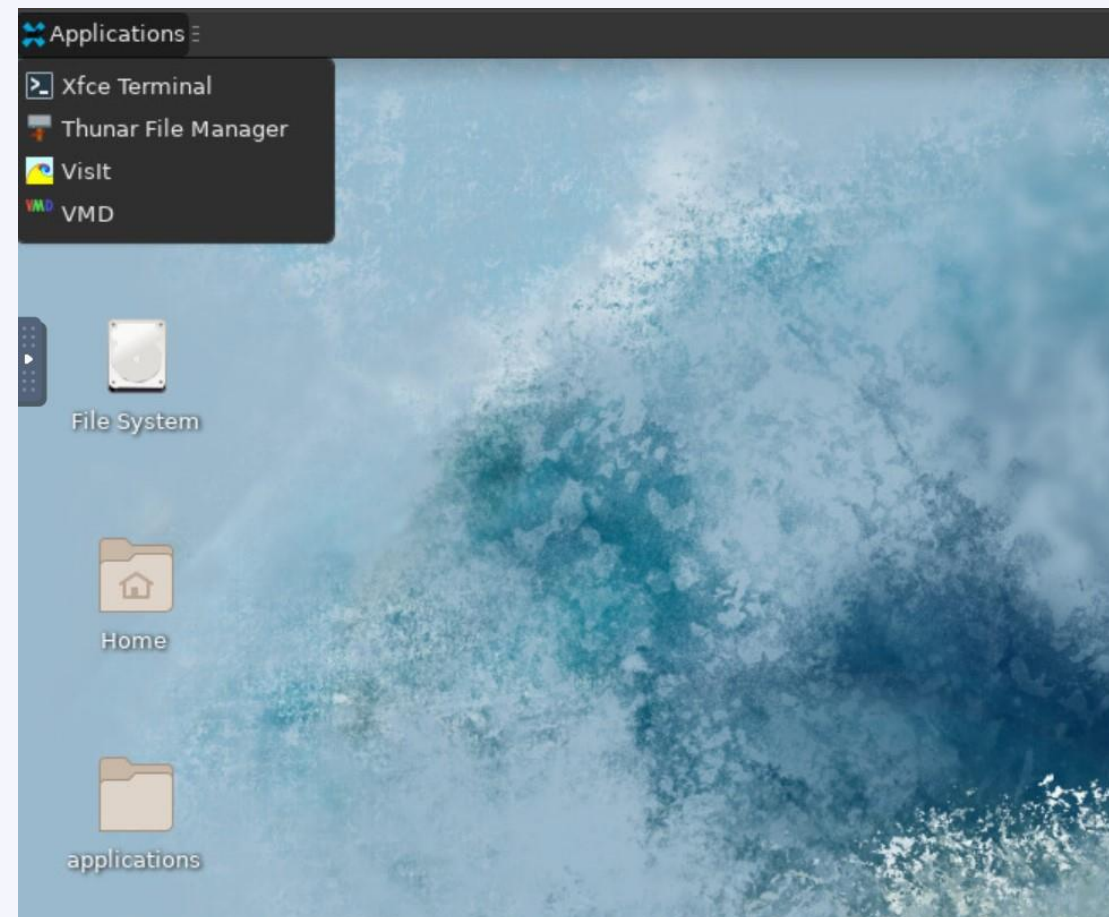
Your new kernel should be available there



Open OnDemand: noVNC Desktop

Desktop

- ✓ Ideal for apps with GUI
- ✓ Load (graphical) modules in Terminal, e.g. MATLAB, Fluent, ncvview, VisIt, ...
- ✓ Minimalistic Terminal (different from login node)
- ✓ Two visualization apps (VisIt and VMD)





Open OnDemand: Other apps



code-server
System Installed App

Start Visual Studio Code server as a job
Develop, deploy and debug your workflow directly on HPC
Use of the interactive partitions is free of charge and is a good choice
Supports many languages + Github integration



RStudio Server
System Installed App

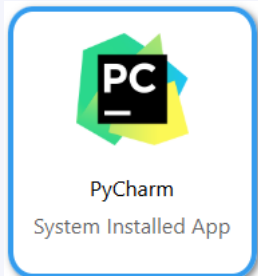
Start RStudio IDE as a job
Install your own packages in VSC_DATA



Tensorboard
System Installed App

Track machine learning metrics and visualize data during the workflow
You need to provide a log folder

Open OnDemand: Other apps



Start PyCharm IDE as a job

Develop, deploy and debug your Python workflow directly on HPC

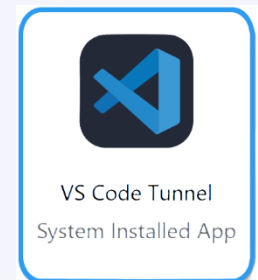
Use Python environment based on Conda or virtual environments



Use Fluent for CFD simulations (single-node)

You need to be in a license group before being able to see the app logo

Contact your supervisor to help you joining an existing license group



Tunnel to the cluster from your local VSCode, so that you code locally, but run remotely! You can enable extensions such as Copilot and more.



Open OnDemand: Resources

E.g. to start a Jupyter notebook

- ✓ Pick a valid Slurm credit account
- ✓ Default partition for most apps: *interactive* (interactive: max 8 cores, 1 GPU instance, 16 hr)
- ✓ Default resources: 1 core, 1 hour, 3400 MB RAM, no GPU

Interactive Apps

Servers

● Interactive Shell

📄 Jupyter Lab

📄 RStudio Server

📄 Tensorboard

📄 code-server

Work in progress

📄 ParaViewWeb -
Work in progress

📄 cryosparc

Jupyter Lab

This app will launch a Jupyter Lab server on one or more nodes.

Account

lp_myproject

Partition

interactive

batch(_long) or bigmem or interactive or gpu or dedicated...

Number of hours

3

Number of cores

1

Required memory per core in megabytes

3400

Number of nodes

1

Number of gpu's

0

[type:] Specify the total number of GPUs slices for the job. An optional GPU type specification can be supplied. For example "A100:3" or "3".

Reservation (optional)

Name of an existing reservation in which the job should run

☐ I would like to receive an email when the session starts

Launch



Software

Software: Centrally installed modules

- ✓ OS: wICE is running on Rocky Linux 9.6
- ✓ Toolchains:
 - *intel* (icc, icpc, ifort; Intel MPI; Intel MKL)
 - *foss* (gcc, g++, gfortran; OpenMPI; ScaLAPACK, OpenBLAS, FFTW)
 - Toolchain year on wICE from 2021a onwards
- ✓ Note: not all toolchains are compatible with each other (conflicting dependencies)
For example, **never mix ...-foss-... and ...-intel-...** modules

Command	Remark
<code>module av</code>	List all available modules
<code>module av Python</code>	List all Python-related modules
<code>module spider Python</code>	Get more info
<code>module load Python/3.12.3-GCCcore-13.3.0</code>	Load a specific module
<code>module list</code>	List all loaded modules and their dependencies
<code>module unload Python/3.12.3-GCCcore-13.3.0</code>	Unload a module (but dependencies still stay)
<code>module purge</code>	Remove all modules from your work session

Software: Your specific needs

- ✓ If you need a software package that is not yet available as a module, you can ask us to provide it
 - ✓ For Python and R packages, however, we prefer that you install it yourself
 - ✓ Read more about [Python Package Management](#)
 - ✓ Read more about [R Package Management](#)
- ✓ If you want to compile software yourself:
 - ✓ Best install it in your VSC_DATA
 - ✓ Use *intel* or *foss* toolchains
 - ✓ Compile your code on a compute node (with interactive job)
 - ✓ If you run into problems, don't hesitate to contact us



Starting to compute

Resource glossary

- ✓ **Cluster:** which machine to use? wICE or Mindwell?
- ✓ **Nodes:** how many compute servers to allocate?
- ✓ **Cores:** how many cores per node to allocate?
- ✓ **Memory:** how much CPU memory to allocate?
- ✓ **Partition:** batch, gpu, bigmem, ...?
- ✓ **Walltime:** how long to allocate the resources?
- ✓ **Storage:** how much storage (data, scratch, etc) will the job require?
- ✓ **Credits:** how many compute credits will the job consume?



Default resources

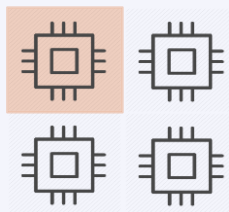
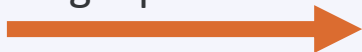
- ✓ **Cluster:** the cluster associated with the node you're on (wICE or Mindwell)
- ✓ **Nodes:** 1
- ✓ **Cores:** 1
- ✓ **Memory:** depends on which node(s) you use
- ✓ **Partition:** batch
- ✓ **Walltime:** typically 1 hour
- ✓ **Storage:** no default
- ✓ **Credits:** no default



Serial application

1 process on 1 core

Single process



One node

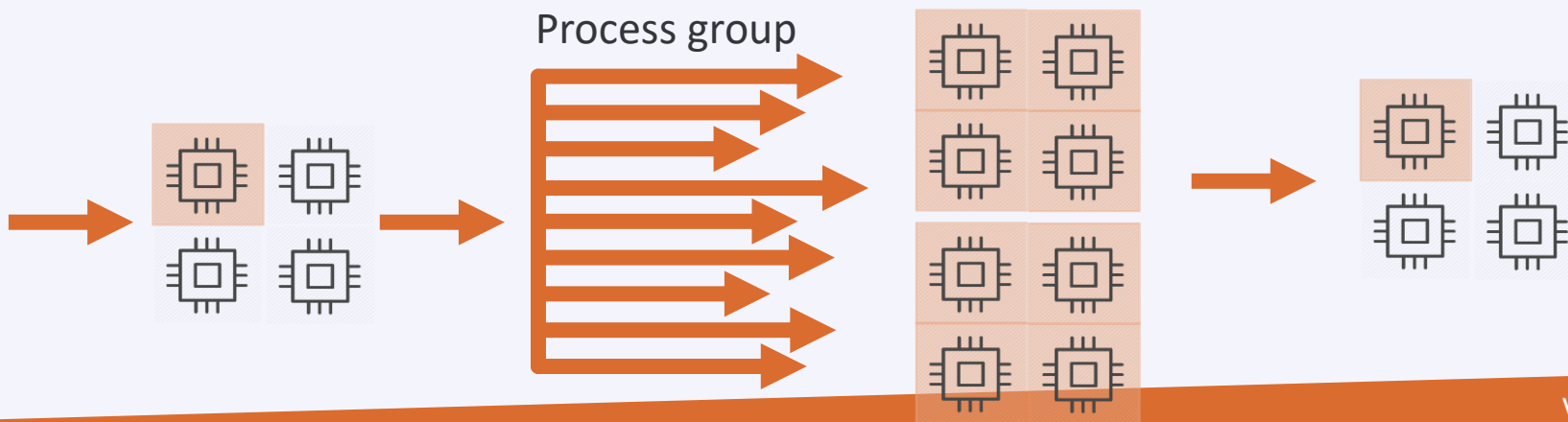
Multi-threaded application

N threads on N cores
on **1 node**

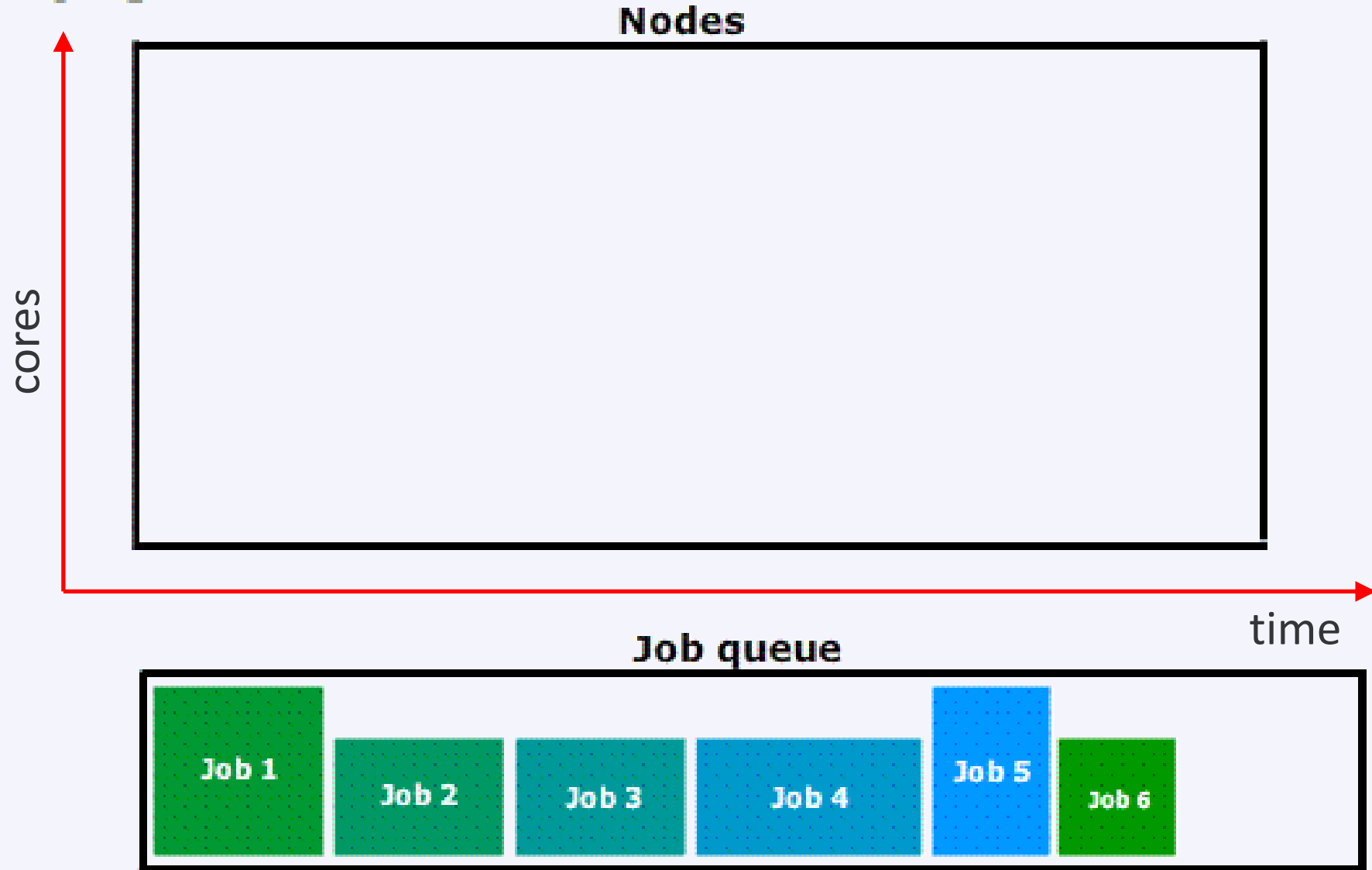


Distributed application

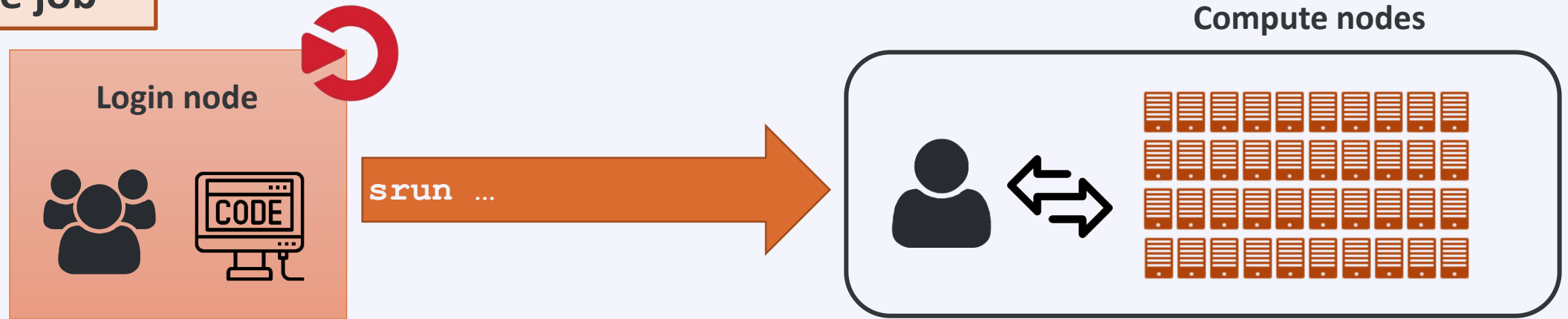
Many processes on
multiple cores,
possibly spread over
multiple nodes



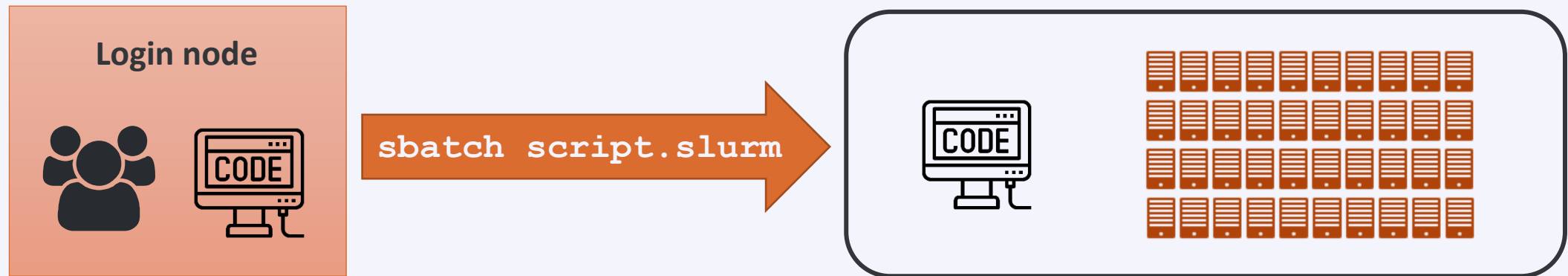
Backfill



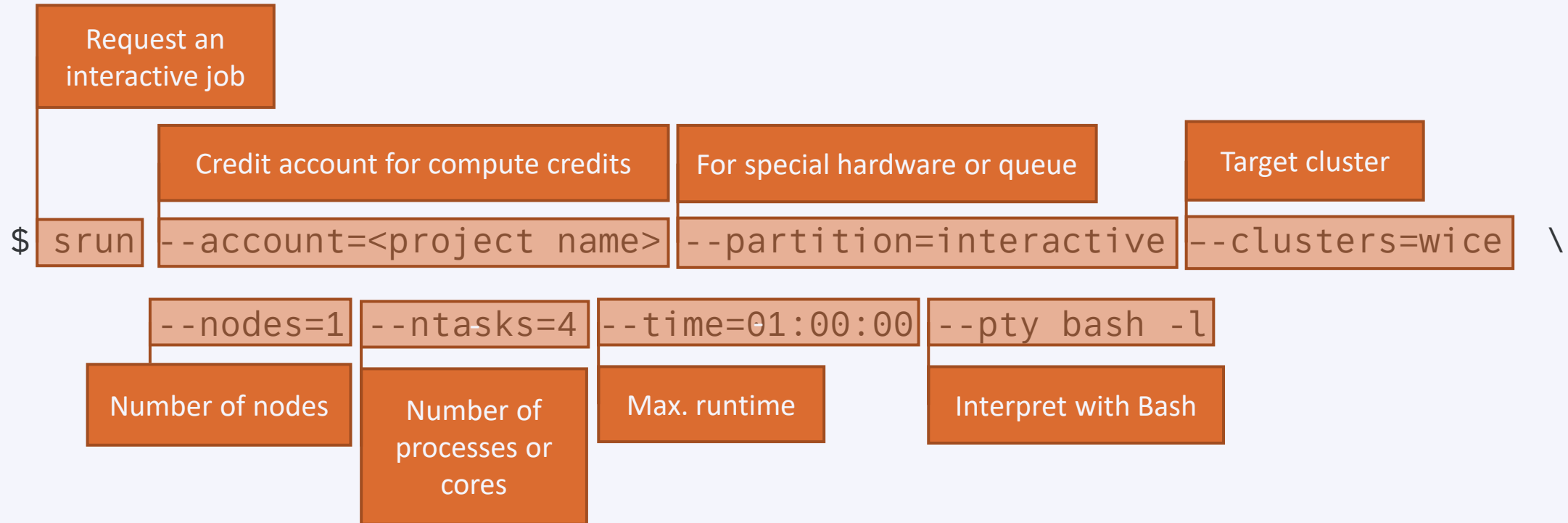
Interactive job



Batch job



Example interactive job on wICE



Remark

- ❑ Specifying <project name> for credits is mandatory, *e.g.* **-A lp_hpcinfo**
- ❑ Implicit defaults for an interactive job on wICE are
`-n 1 -t 01:00:00 -p batch --mem-per-cpu=3400M`

Slurm command-line tools

Command	Purpose
\$ sbatch ...	Submit a batch job
\$ srun ...	Submit an interactive job
\$ scancel --cluster=wice <JobID>	Cancel a specific pending or running job
\$ scontrol show job --clusters=wice <JobID> \$ slurm_jobinfo <JobID>	Detailed job info (very useful to diagnose issues)
\$ squeue --clusters=wice --me --long	Status of all recent jobs
\$ squeue --clusters=wice --me --start	Give a rough estimate of when your jobs will start
\$ sinfo --clusters=wice	Info about the state of available partitions and nodes
\$ sacct --clusters=wice --batch --job <JobID>	Show the jobscript used for a given batch job
\$ slurmtop	Overview of the cluster
\$ sam-balance	Overview of all your available credit projects
\$ sam-list-allocations	Detailed overview of your credit allocation history

Slurm job submit options

Slurm option	Remarks
<code>--nodes=X --ntasks-per-node=Y</code> <code>--ntasks=X*Y</code> <code>--ntasks=X --cpus-per-task=Y</code>	or or (e.g. Hybrid MPI + OpenMP)
<code>--partition=<partition_name></code>	Default: "batch" partition
<code>--mem-per-cpu=<size><M></code>	Min memory per core, e.g. 3400M (wICE)
<code>--time=<dd-hh:mm:ss></code>	e.g. 1-12:30:00
<code>--job-name=<job_name></code>	
<code>--output=<file_template></code>	STDOUT; default="slurm-%j.out"
<code>--error=<file_template></code>	Default: redirect to STDOUT
<code>--mail-type=FAIL,BEGIN,END</code>	
<code>--mail-user=<email-address></code>	
<code>--export=<ALL, key=value></code>	Additional variables to pass to the job
<code>--account=<account_name></code>	Credit account (mandatory)

Slurm job submit options

Slurm option	Remarks
<code>--account=<account_name></code>	Mandatory (credits)
<code>--partition=<partition_name></code>	Default: "batch" partition
<code>--time=<dd-hh:mm:ss></code>	e.g. 1-12:30:00
<code>--nodes=X --ntasks-per-node=Y</code> <code>--ntasks=X*Y</code> <code>--ntasks=X --cpus-per-task=Y</code>	or or (e.g. Hybrid MPI + OpenMP)
<code>--mem-per-cpu=<size><M></code>	Min memory per core, e.g. 3400M (wICE)
<code>--job-name=<job_name></code>	
<code>--output=<file_template></code>	STDOUT+STDERR; default="slurm-%j.out"
<code>--mail-type=FAIL,BEGIN,END</code>	
<code>--mail-user=<email-address></code>	
<code>--export=<ALL,key=value></code>	Additional variables to pass to the job

Long and short options

Some (not all) of the `sbatch`, `srun` and `salloc` command line options also have a short version

Short version	Long version	Meaning
-A	--account	Slurm account name
-a	--array	Job array range
-M	--clusters	Cluster name
-c	--cpus-per-task	Number of cores per task (default: 1)
-d	--dependency	Job dependency (after, afterok, afterany, ...)
-i	--input	STDIN filename
-e	--error	STDERR filename
-o	--output	STDOUT filename
-t	--time	Maximum walltime
-N	--nodes	Minimum num nodes
-n	--ntasks	Maximum num tasks
-p	--partition	Partition name

Interactive jobs

- ✔ Interactive job with default options (1 core for 1 hour on the *batch* partition)

```
$ srun --clusters=wice --account=lp_hpcinfo --pty bash -l
```

- ✔ Interactive job with X-forwarding

```
$ srun --clusters=wice --account=lp_hpcinfo --x11 --pty bash -l
```

- ✔ Interactive job with multiple cores, on the *interactive* partition

```
$ srun --clusters=wice --account=lp_hpcinfo --nodes=1 --ntasks=4 \
    --partition=interactive --pty bash -l
```

- ✔ Interactive job with multiple cores and a GPU device

```
$ srun --clusters=wice --account=lp_hpcinfo --nodes=1 --ntasks=18 \
    --partition=gpu --gpus-per-node=1 --pty bash -l
```

- ✔ To join one of your running jobs (connecting to the first compute node in the job's allocation):

```
$ srun -M wice --jobid=<JobID> --overlap -pty bash -l
```

Example Slurm job script

jobscript.slurm

```
#!/bin/bash -l
#SBATCH --clusters=wise
#SBATCH --account=lp_hpcinfo
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --mem-per-cpu=3400M
#SBATCH --job-name=hpc_workflow
#SBATCH --mail-type=BEGIN,END,FAIL
#SBATCH --mail-user=my.name@kuleuven.be
```

Shebang

Resource list

```
module load intel/2024a
module load Python/3.12.3-GCCcore-13.3.0
which python
cd $VSC_SCRATCH/projects/simulations
cp -r $VSC_DATA/input_data .
```

Loading modules

Move data

```
python modelling.py
```

Execute commands

```
cp -r output_data $VSC_DATA
rm -rf ./input_data ./output_data
```

Move data

Batch job workflow

- ✓ Submit the job to the scheduler
- ✓ Receive a unique JobID
- ✓ Error and output files

Job script

```
#!/bin/bash -l  
...
```

Submit command

```
$ sbatch simulation.slurm
```

JobID

Submitted batch job 60042478 on cluster wice

stderr, stdout

```
$ ls *.out  
slurm-60042478.out
```

Output files

- ✓ STDERR and STDOUT are by default redirected to a single file:
`slurm-<JobID>.out`
- ✓ Contains job info, all errors and warnings, and printouts
- ✓ Worth going through (especially if it's a new kind of job)
- ✓ Address all warnings and errors (if you can)
- ✓ Typical error examples on the right

STDOUT + STDERR

```
$ ls slurm-*.out  
slurm-60042478.out
```

Out of memory

```
slurmstepd: error: Detected 1 oom-kill event(s).  
Some of your processes may have been killed by the  
cgroup out-of-memory handler.
```

Short walltime

```
slurmstepd: error: *** JOB 60042478 ON s28c11n2  
CANCELLED AT 2023-02-08T10:03:43 DUE TO TIME LIMIT  
***
```

Low disk space

```
IOError: [Errno 122] Disk quota exceeded
```

Output files

- ✓ Standard output and error channels can be redirected to other files:

```
#SBATCH --output ...
```

```
#SBATCH --error ...
```

stdout

```
$ ls slurm-*.out  
slurm-60041238.out
```

Output file

```
SLURM_JOB_ID: 60033947  
SLURM_JOB_USER: vscXXXXX  
SLURM_JOB_ACCOUNT: lp_wice_pilot  
SLURM_JOB_NAME: testjob  
SLURM_CLUSTER_NAME: wice  
SLURM_JOB_PARTITION: batch  
SLURM_NNODES: 1  
SLURM_NODELIST: m28c30n4  
SLURM_JOB_CPUS_PER_NODE: 72  
Date: Tue Jan 10 17:02:04 CET 2023  
Walltime: 00-01:00:00  
=====  
/apps/leuven/rocky8/icelake/2021a/  
softwar  
e/intel-compilers/2021.2.0/compile  
r/2021.2.0/linux/bin/intel64/icc  
cp: cannot stat '/apps/leuven/trai  
ning/test': No such file or directo  
ry  
Hello World
```

Resource
summary

stdout

Job monitoring

Example

```
$ scontrol show job --clusters=wice 60049330
  JobId=60049330  JobName=f70.slurm
  UserId=vsc3...(253...)  GroupId=vsc3...(253...)
Account=lp_my_project  QOS=lp_my_project
JobState=PENDING Reason=QOSGrpBillingMinutes
  SubmitTime=2023-02-16T00:24:58  EligibleTime=2023-02-16T00:24:58
  Partition=batch  NodeList=  NumNodes=4-4  NumCPUs=288  NumTasks=288  CPUs/Task=1
  TRES=cpu=288,mem=979200M,node=4,billing=733
  MinCPUsNode=72  MinMemoryCPU=3400M
  WorkDir=/vsc-hard-mounts/leuven-data/3../vsc3.../Odonate_SOM
  StdErr=/vsc-hard-mounts/leuven-data/3../vsc3.../Odonate_SOM/slurm-60049330.out
  StdIn=/dev/null
  StdOut=/vsc-hard-mounts/leuven-data/3../vsc3.../Odonate_SOM/slurm-60049330.out
```

Diagnosis

Why does my job remain in a pending state?

Check out the “Reason” on Slurm docs: https://slurm.schedmd.com/resource_limits.html

Other wICE partitions

GPU

```
#SBATCH --partition=gpu
#SBATCH --nodes=1
#SBATCH --ntasks=18
#SBATCH --gpus-per-node=1
```

Big memory

```
#SBATCH --partition=bigmem
#SBATCH --nodes=1
#SBATCH --tasks-per-node=72
#SBATCH --mem-per-cpu=28000M
```

Interactive partitions

- ✓ Accessible via command line and Open OnDemand
- ✓ To quickly compile, test, debug your (parallel) application
- ✓ To pre- or postprocess your data (incl. visualizations)
- ✓ Short queue time
- ✓ 4 nodes on wICE: 64 cores/node, 512 GiB mem, 1 physical GPU (split in 7 instances)
- ✓ 2 nodes on Mindwell: 192 cores/node, 768 GiB mem, 2 Physical GPUs
- ✓ Max resource per user: 8 cores, 1 GPU instance, 16 hour walltime
- ✓ Free of charge

Interactive job

```
$ srun --account=lp_myproject --clusters=wice \  
    --partition=interactive --nodes=1 --time=16:00:00 \  
    --gpus-per-node=1 --pty bash -l
```

Debugging partitions

- ✓ Accessible via command line and Open OnDemand
- ✓ Quickly test if your (parallel) application works
- ✓ Short queue time
- ✓ Only one job at a time, max walltime: 1 hr

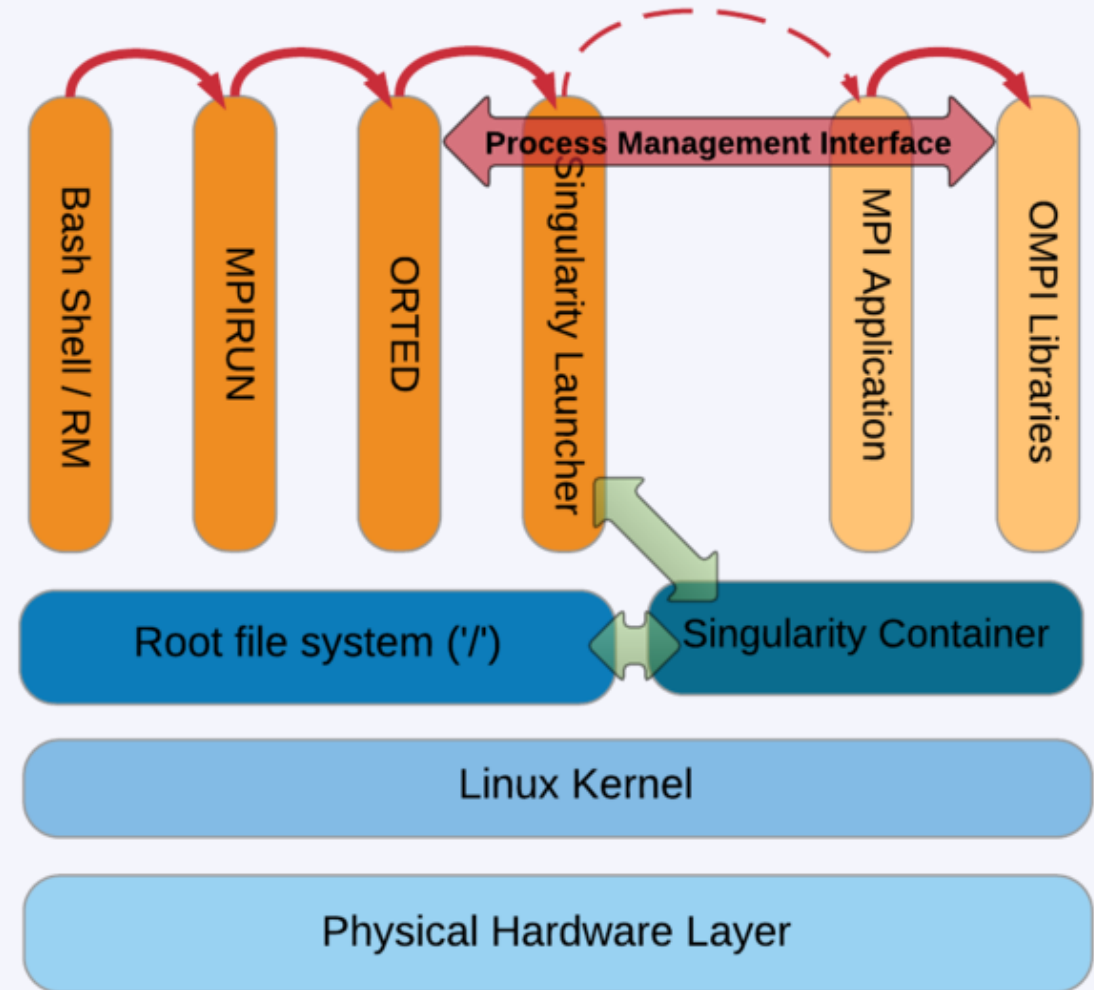
Cluster	Partition	Resources
wICE	interactive	4 Icelake nodes, each with 7 MIG instances of 1 A100 GPU
	gpu_a100_debug	1 Icelake node with 1 A100 GPU
Mindwell	interactive	2 Granite Rapids nodes, each with 2 RTX5000 GPUs

Containers: Apptainer (Singularity)

- ✓ What?
 - Self-contained OS image in which you can install software and include data
- ✓ Why?
 - Package managers like apt can be convenient
 - Portability: images can be used on various hardware (but performance may be sub-par)
- ✓ How?
 - Create the image
 - Run it on wICE or Mindwell
 - Parallelization should work (multi-node MPI runs may require some care)

```
#!/bin/bash -l
#SBATCH --time=30:00
#SBATCH --nodes=1
#SBATCH --ntasks=72
#SBATCH --clusters=wice
#SBATCH --account=lp_hpcinfo
apptainer run Project.sif ./model.exe
```

script.slurm



High-throughput computing with *atools*

- ✓ You can use `atools` to handle high-throughput computing workloads
- ✓ `module load atools/<version>`
- ✓ VSC training: Workflows for Data Science
- ✓ Training material: <https://github.com/gjbex/Workflows-for-HPC>
- ✓ Recording: <https://www.youtube.com/watch?v=GlwaRyi1LVA>
- ✓ User documentation: <https://atools.readthedocs.io/en/latest/>



Demo: try things yourself

Demo: try things yourself

- ✓ Request to become a member of the *lp_hpcinfo* group via account.vscentrum.be
- ✓ Transfer a file with FileZilla
- ✓ Check your disk quota
- ✓ Check your available credits
- ✓ Check/load/list/unload/purge module

Demo: try things yourself

- ✓ Copy `/apps/leuven/training/HPC_intro/` to your `$VSC_DATA`
- ✓ Submit `cpujob.slurm` to the cluster
- ✓ List all your jobs with `squeue -M wice`
- ✓ Check the information about the job with `slurm_jobinfo -M wice <job_ID>`
- ✓ Modify the `mpi.slurm` script to request 1 node, 72 cores for 30 minutes, with job start/end e-mail notifications
- ✓ Check the status of all the jobs

Demo: job monitoring

- ✓ Submit an interactive job
 - Run your program on a compute node
 - Open a new terminal and `ssh` to the allocated compute node
 - Check the resources usage with `htop`
- ✓ Submit a GPU batch job
 - While the job is running, check its details with `slurm_jobinfo`
 - and check resource on the compute node using `ssh`, `htop` and `nvidia-smi`

Useful links

Helpdesk (email): hpcinfo@kuleuven.be

Helpdesk (online form): [https://admin.kuleuven.be/icts/HPInfo form/HPC-info-formulier](https://admin.kuleuven.be/icts/HPInfo_form/HPC-info-formulier)

VSC website: <https://www.vscentrum.be>

VSC documentation: <https://docs.vscentrum.be>

VSC agenda, training sessions, events (User Day): <https://www.vscentrum.be/vsctraining>

System status page: <https://status.vscentrum.be>

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